

Material Science Engineering

By Dr. Dinesh Shringi

Syllabus

- ME 202 A: MATERIAL TECHNOLOGY (M) 3L 3 Hours, 60 Marks
- Crystalline Nature of Solids: Crystal structure and space lattice, packing arrangement, coordination number, crystallographic planes and Miller indices;
- Imperfection in crystals, point defects – vacancies, Schottky, Frenkel – and line defects – dislocations, edge, screw; Burgers vector;
- Diffusion and crystallization of metals, Cooling curves, Nucleation, Grain growth.
- Plastics Deformation of Metals and Alloys: Internal stresses, role of dislocation, slip, Frank-Read source, twinning; Strain hardening,
- Effect on mechanical properties – Stress-strain curve, Strength and Toughness, Hardness, Impact, Ductility and Brittleness, Fatigue, Creep, and Fracture. Effect of temperature – Recovery, Recrystallisation, Hot and Cold deformation.
- Phase Diagrams and Structure of Alloys (Iron-Carbon system): Components and Phases, Solidification and Solubility; Equilibrium Phase Diagrams, Dendritic segregation (Coring), Diffusion, Precipitation hardening, Ageing; Eutectics and Eutectoids.
- Heat Treatment of Steels: Phase transformation, TTT diagrams (S curves); Stress relief and Annealing, Quenching and Hardening, Tempering, Normalizing, Hardenability, Case hardening, Carburising, Nitriding, Cyaniding.
- Engineering Materials: Effects of alloying elements in steels; Plain Carbon steels, Alloy steels –Stainless steels, Magnetic materials; Non-ferrous metals and alloys – Brass and Bronze, Aluminium alloys, Bearing materials.
- Brief introduction to non-metallic materials: Polymers, Ceramics, Composites, Nano materials, Smart materials and their applications.
- Powder metallurgy: Metallic powders – Production and Characteristics; Compaction and Sintering; Materials and Products.

Introduction

- Materials are probably more deep seated in our culture than most of us realize.
- Transportation, housing, clothing, communication, recreation, and food production— virtually every segment of our everyday lives is influenced to one degree or another by materials.
- The development of many technologies that make our existence so comfortable has been intimately associated with the accessibility of suitable materials.

WHY STUDY MATERIALS SCIENCE

- Many an applied scientist or engineer, whether mechanical, civil, chemical, or electrical, is at one time or another exposed to a design problem involving materials, such as a transmission gear, the superstructure for a building, an oil refinery component, or an integrated circuit chip. Of course, materials scientists and engineers are specialists who are totally involved in the investigation and design of materials.
- Many times, a materials problem is one of selecting the right material from the thousands available. The final decision is normally based on several criteria. First, the in-service conditions must be characterized, for these dictate the properties required of the material. On only rare occasions does a material possess the maximum or ideal combination of properties. Thus, it may be necessary to trade one characteristic for another.
- The classic example involves strength and ductility; normally, a material having a high strength has only a limited ductility. In such cases, a reasonable compromise between two or more properties may be necessary.
- A second selection consideration is any deterioration of material properties that may occur during service operation. For example, significant reductions in mechanical strength may result from exposure to elevated temperatures or corrosive environments.
- Finally, probably the overriding consideration is that of economics: What will the finished product cost? A material may be found that has the ideal set of properties but is prohibitively expensive. Here again, some compromise is inevitable. The cost of a finished piece also includes any expense incurred during fabrication to produce the desired shape.

CLASSIFICATION OF MATERIALS

- Solid materials have been conveniently grouped into three basic categories: metals, ceramics, and polymers, a scheme based primarily on chemical makeup and atomic structure.
- Another category is advanced materials—those used in high-technology applications, such as semiconductors, biomaterials, smart materials, and nanoengineered materials;

Metals

— Platinum
—  Silver
—  Copper
—  Iron/Steel
— Titanium
— Aluminum
— Magnesium

Ceramics

— ZrO_2
—  Al_2O_3
— SiC, Si_3N_4
—  Glass
—  Concrete

Composites

  GFRP
  CFRP
 Woods

Polymers

— PTFE
 PVC
—  PS
—  PE
—  Rubber

ADVANCED MATERIALS

- Materials utilized in high-technology (or high-tech) applications are sometimes termed *advanced materials*.
- These advanced materials are typically traditional materials whose properties have been enhanced and also newly developed, high-performance materials. Furthermore, they may be of all material types (e.g., metals, ceramics, polymers) and are normally expensive.

ADVANCED MATERIALS :Smart Materials

- **Semiconductors**

- *Semiconductors* have electrical properties that are intermediate between those of electrical conductors (i.e., metals and metal alloys) and insulators (i.e., ceramics and polymers). Furthermore, the electrical characteristics of these materials are extremely sensitive to the presence of minute concentrations of impurity atoms, for which the concentrations may be controlled over very small spatial regions. Semiconductors have made possible the advent of integrated circuitry that has totally revolutionized the electronics and computer industries (not to mention our lives) over the past three decades.

- **Biomaterials**

- *Biomaterials* are employed in components implanted into the human body to replace diseased or damaged body parts. These materials must not produce toxic substances and must be compatible with body tissues (i.e., must not cause adverse biological reactions). All of the preceding materials—metals, ceramics, polymers, composites, and semiconductors— may be used as biomaterials.

ADVANCED MATERIALS : Smart Materials

- *Smart (or intelligent) materials* are a group of new and state-of-the-art materials now being developed that will have a significant influence on many of our technologies. The adjective *smart* implies that these materials are able to sense changes in their environment and then respond to these changes in predetermined manners—traits that are also found in living organisms. In addition, this *smart* concept is being extended to rather sophisticated systems that consist of both smart and traditional materials.
- Components of a smart material (or system) include some type of sensor (which detects an input signal) and an actuator (which performs a responsive and adaptive function). Actuators may be called upon to change shape, position, natural frequency, or mechanical characteristics in response to changes in temperature, electric fields, and/or magnetic fields.
- Four types of materials are commonly used for actuators: shape-memory alloys, piezoelectric ceramics, magnetostrictive materials, and electrorheological/magnetorheological fluids.
- *Shape-memory alloys* are metals that, after having been deformed, revert to their original shape when temperature is changed
- *Piezoelectric ceramics* expand and contract in response to an applied electric field (or voltage); conversely, they also generate an electric field when their dimensions are altered .
- The behavior of *magnetostrictive materials* is analogous to that of the piezoelectrics, except that they are responsive to magnetic fields.
- Also, *electrorheological* and *magnetorheological fluids* are liquids that experience dramatic changes in viscosity upon the application of electric and magnetic fields, respectively.

Nanomaterials

- One new material class that has fascinating properties and tremendous technological promise is the *nanomaterials*, which may be any one of the four basic types—metals, ceramics, polymers, or composites. However, unlike these other materials, they are not distinguished on the basis of their chemistry but rather their size; the *nano* prefix denotes that the dimensions of these structural entities are on the order of a nanometer (10^{-9} m)—as a rule, less than 100 nanometers (nm; equivalent to the diameter of approximately 500 atoms).
- Prior to the advent of nanomaterials, the general procedure scientists used to understand the chemistry and physics of materials was to begin by studying large and complex structures and then investigate the fundamental building blocks of these structures that are smaller and simpler. This approach is sometimes termed *top-down science*.
- However, with the development of scanning probe microscopes, which permit observation of individual atoms and molecules, it has become possible to design and build new structures from their atomic-level constituents, one atom or molecule at a time (i.e., “materials by design”). This ability to arrange atoms carefully provides opportunities to develop mechanical, electrical, magnetic, and other properties that are not otherwise possible. We call this the *bottom-up approach*, and the study of the properties of these materials is termed *nanotechnology*.
- Some of the physical and chemical characteristics exhibited by matter may experience dramatic changes as particle size approaches atomic dimensions. For example, materials that are opaque in the macroscopic domain may become transparent on the nanoscale; some solids become liquids, chemically stable materials become combustible, and electrical insulators become conductors. Furthermore, properties may depend on size in this nanoscale domain. Some of these effects are quantum mechanical in origin, whereas others are related to *surface phenomena*—the proportion of atoms located on surface sites of a particle increases dramatically as its size decreases. Because of these unique and unusual properties, nanomaterials are finding niches in electronic, biomedical, sporting, energy production, and other industrial applications.

Lecture 1 Crystalline Nature of Solids

INTRODUCTION

- Some of the important properties of solid materials depend on geometric atomic arrangements and also the interactions that exist among constituent atoms or molecules.
- The present discussion is devoted to the structure of materials, specifically, to some of the arrangements that may be assumed by atoms in the solid state.
- Within this framework, concepts of crystallinity and noncrystallinity are introduced. For crystalline solids, the notion of crystal structure is presented, specified in terms of a unit cell.
- The three common crystal structures found in metals are then detailed, along with the scheme by which crystallographic points, directions, and planes are expressed.

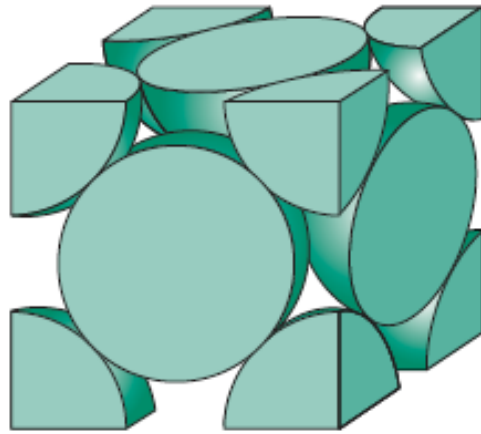
FUNDAMENTAL CONCEPTS

- A **crystalline** material is one in which the atoms are situated in a repeating or periodic array over large atomic distances—that is, long-range order exists, such that upon solidification, the atoms will position themselves in a repetitive three-dimensional pattern, in which each atom is bonded to its nearest neighbor atoms.
- All metals, many ceramic materials, and certain polymers form crystalline structures under normal solidification conditions.
- For those that do not crystallize, this long-range atomic order is absent; these *noncrystalline* or *amorphous* materials are discussed briefly at the end.

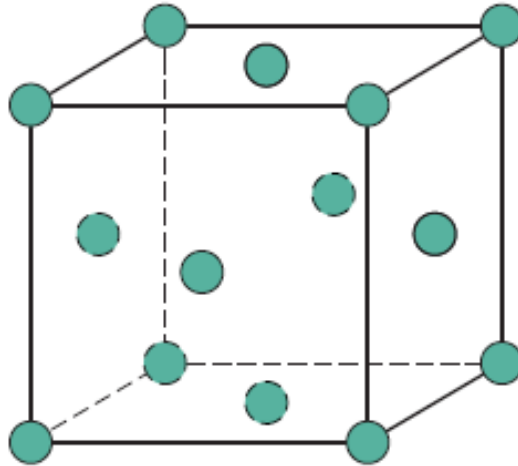
crystal structure & Lattice

- Some of the properties of crystalline solids depend on the **crystal structure** of the material, the manner in which atoms, ions, or molecules are spatially arranged.
- There is an extremely large number of different crystal structures all having long-range atomic order; these vary from relatively simple structures for metals to exceedingly complex ones, as displayed by some of the ceramic and polymeric materials.
- When crystalline structures are described, atoms (or ions) are thought of as being solid spheres having well-defined diameters. This is termed the *atomic hard-sphere model* in which spheres representing nearest-neighbor atoms touch one another. An example of the hard-sphere model for the atomic arrangement found in some of the common elemental metals is displayed in Figure(Next Slide) .
- In this particular case all the atoms are identical. Sometimes the term **lattice** is used in the context of crystal structures; in this sense *lattice* means a three-dimensional array of points coinciding with atom positions (or sphere centers).

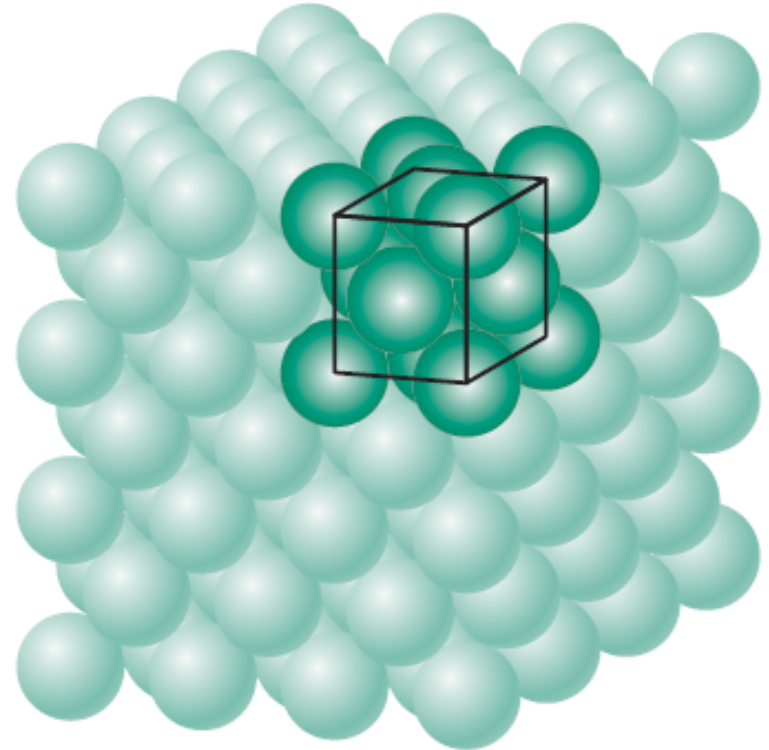
For the face-centered cubic crystal structure, (a) a hard-sphere unit cell representation, (b) a reduced sphere unit cell, and (c) an aggregate of many atoms.
[Figure (c) adapted from W. G. Moffatt, G. W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. I, *Structure*,



(a)



(b)



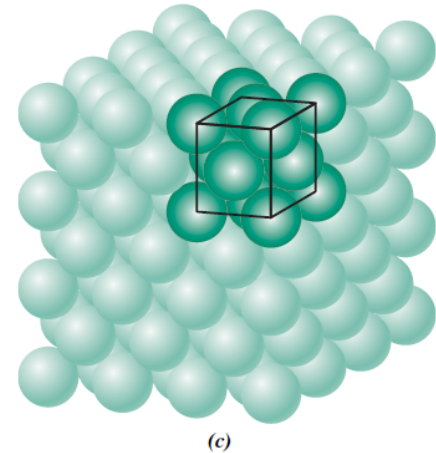
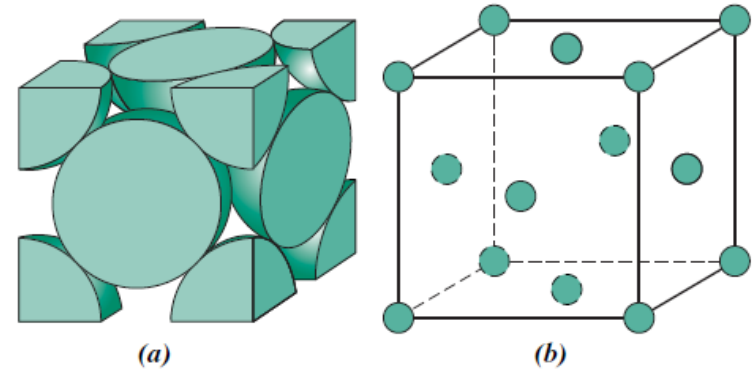
(c)

UNIT CELLS

- The atomic order in crystalline solids indicates that small groups of atoms form a repetitive pattern.
- Thus, in describing crystal structures, it is often convenient to subdivide the structure into small repeat entities called **unit cells**.
- Unit cells for most crystal structures are parallelepipeds or prisms having three sets of parallel faces; one is drawn within the aggregate of spheres (Figure earlier c), which in this case happens to be a cube.
- A unit cell is chosen to represent the symmetry of the crystal structure, wherein all the atom positions in the crystal may be generated by translations of the unit cell integral distances along each of its edges.
- Thus, the unit cell is the basic structural unit or building block of the crystal structure and defines the crystal structure by virtue of its geometry and the atom positions within.
- Convenience usually dictates that parallelepiped corners coincide with centers of the hard-sphere atoms. Furthermore, more than a single unit cell may be chosen for a particular crystal structure; however, we generally use the unit cell having the highest level of geometrical symmetry.

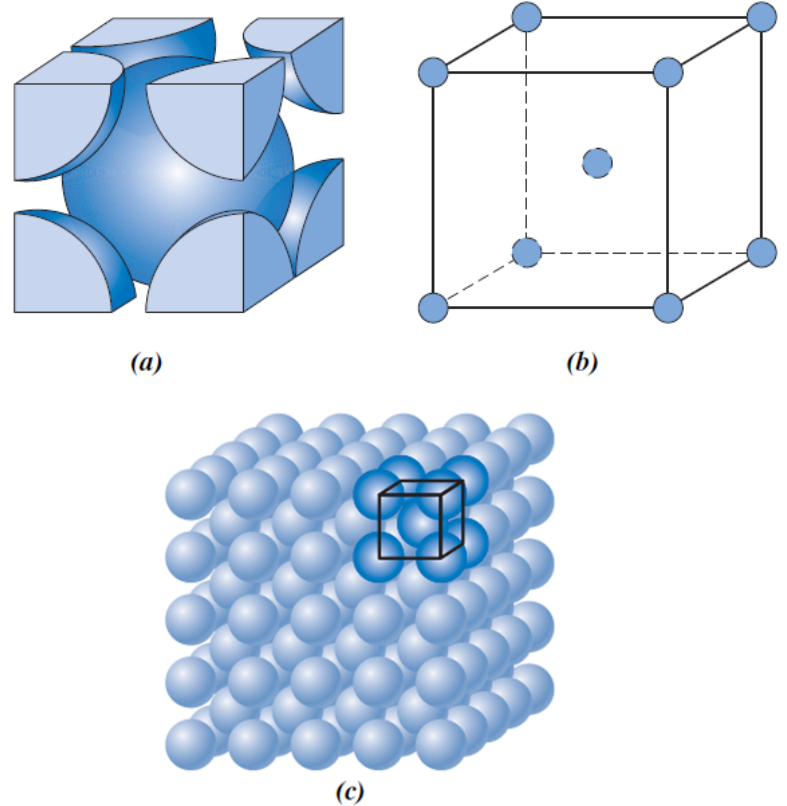
The Face-Centered Cubic Crystal Structure

- The crystal structure found for many metals has a unit cell of cubic geometry, with atoms located at each of the corners and the centers of all the cube faces.
- It is aptly called the **face-centered cubic (FCC)** crystal structure.
- Some of the familiar metals having this crystal structure are copper, aluminum, silver, and gold .
- Figure *a* shows a hard-sphere model for the FCC unit cell, whereas in Figure *b* the atom centers are represented by small circles to provide a better perspective on atom positions.
- The aggregate of atoms in Figure *c* represents a section of crystal consisting of many
- FCC unit cells. These spheres or ion cores touch one another across a face diagonal;



The Body-Centered Cubic Crystal Structure

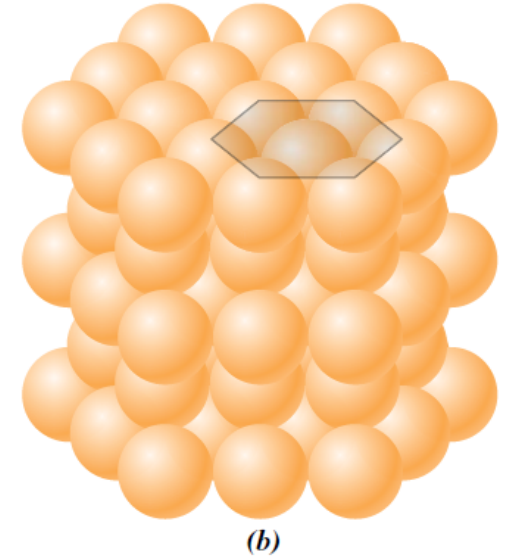
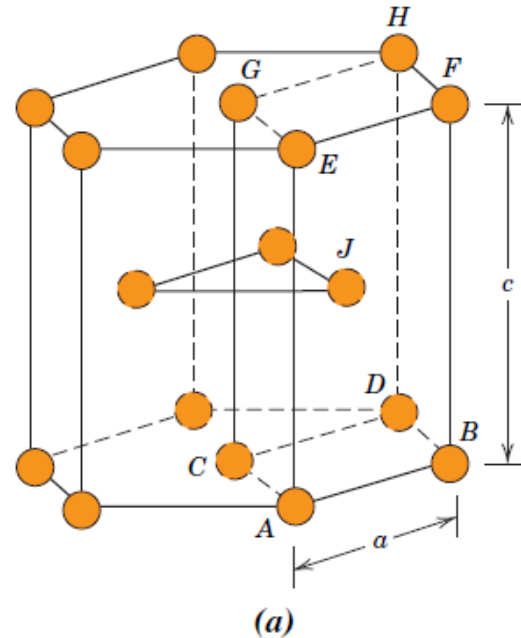
- Another common metallic crystal structure also has a cubic unit cell with atoms located at all eight corners and a single atom at the cube center.
- This is called a body-centered cubic (BCC) crystal structure. A collection of spheres depicting this crystal structure is
- shown in Figure c, whereas Figures a and b are diagrams of BCC unit cells with the atoms represented by hard-sphere and reduced-sphere models, respectively.
- Chromium, iron, tungsten, and several other metals exhibit a BCC structure.
- Each BCC unit cell has eight corner atoms and a single center atom, which is wholly contained within its cell



For the body-centered cubic crystal structure, (a) a hard-sphere unit cell representation, (b) a reduced-sphere unit cell, and (c) an aggregate of many atoms. [Figure (c) from W. G. Moffatt, G. W. Pearsall, and J.

The Hexagonal Close-Packed Crystal Structure

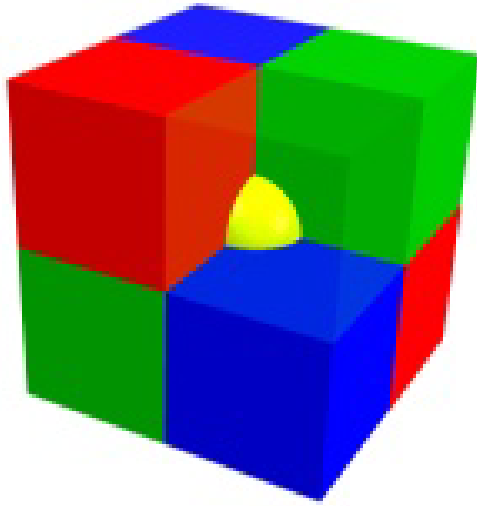
- Not all metals have unit cells with cubic symmetry; the final common metallic crystal structure to be discussed has a unit cell that is hexagonal.
- Figure *a* shows a reduced sphere unit cell for this structure, which is termed **hexagonal close-packed (HCP)**;
- An assemblage of several HCP unit cells is presented in Figure *b*. The top and bottom faces of the unit cell consist of six atoms that form regular hexagons and surround a single atom in the center.
- Another plane that provides three additional atoms to the unit cell is situated between the top and bottom planes. The atoms in this midplane have as nearest neighbors atoms in both of the adjacent two planes.



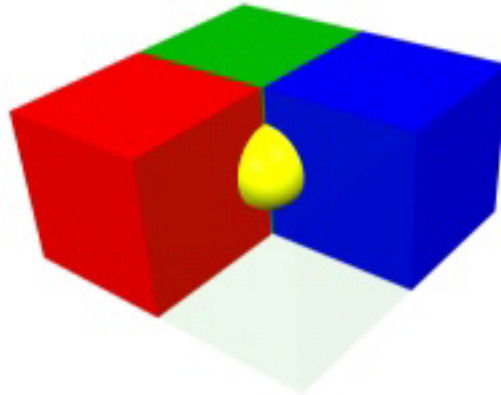
For the hexagonal close-packed crystal structure, (a) a reduced-sphere unit cell (a and c represent the short and long edge lengths, respectively), and (b) an aggregate of many atoms.

Contributions per Unit Cell

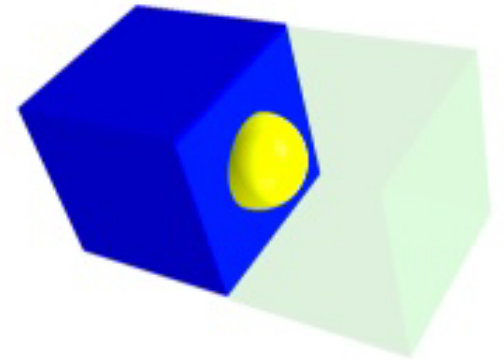
- The particles on the eight corners combine to contribute 1 particle to a unit cell. Particles in the six face centers combine to contribute 3 particles to a unit cell. Those on twelve edges combine to contribute 3 particles per unit cell.



A particle on a **corner** is shared by eight unit cells, so it contributes $\frac{1}{8}$ particle to each.



A particle on an edge is shared by four unit cells, so it contributes $\frac{1}{4}$ particle to each.



A particle on a face center is shared by two unit cells, so it contributes $\frac{1}{2}$ particle to each.

- Atoms that are entirely within a unit cell each contribute a full atom to the cell stoichiometry, but the atoms in the faces, corners, and edges are shared by more than one unit cell, and the number of atoms they contribute to each unit cell is given below

- $\frac{8 \text{ corners}}{\text{unit cell}} \times \frac{1 \text{ atom}}{8 \text{ Corners}} = \frac{1 \text{ atom}}{\text{unit Cell}}$

- Atoms located on the *corners* of the unit cell contribute 1 atom per unit cell.

- $\frac{6 \text{ faces}}{\text{unit cell}} \times \frac{1 \text{ atom}}{2 \text{ Faces}} = \frac{3 \text{ atoms}}{\text{unit Cell}}$

- Atoms located on the *faces* of the unit cell contribute 3 atoms per unit cell.

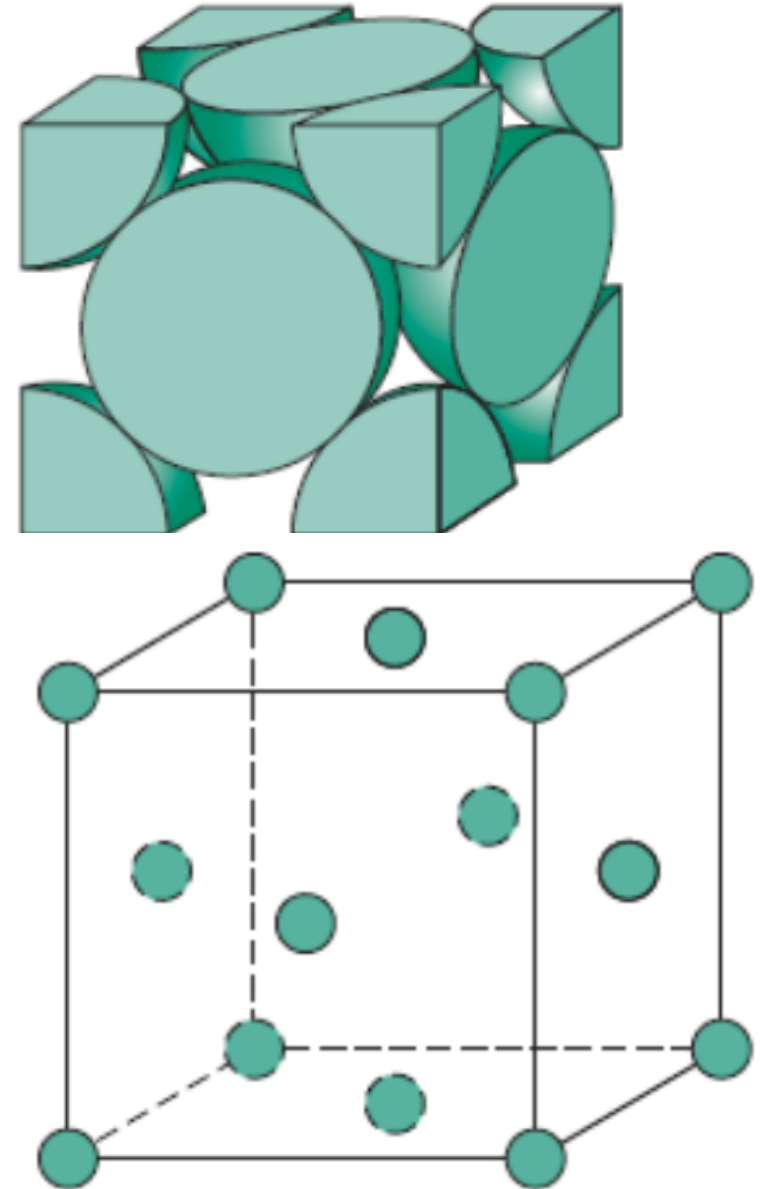
- $\frac{12 \text{ edges}}{\text{unit cell}} \times \frac{1 \text{ atom}}{4 \text{ edges}} = \frac{3 \text{ atoms}}{\text{unit Cell}}$

- Atoms located on the *edges* of the unit cell contribute 3 atoms per unit cell.

No of atoms in Unit Cell

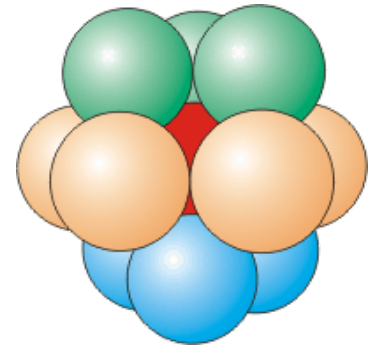
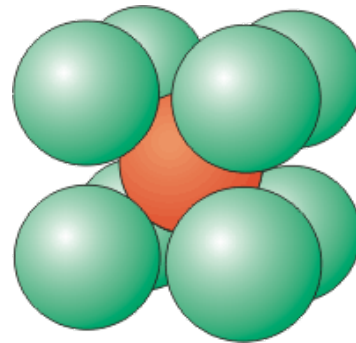
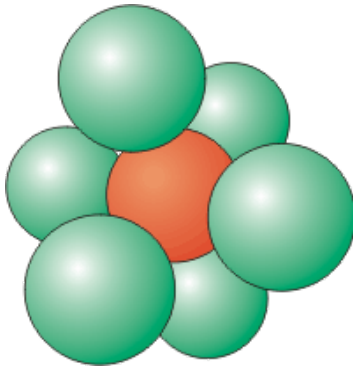
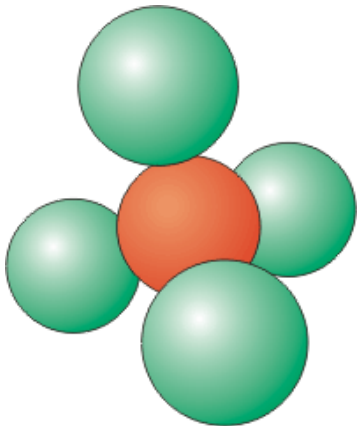
- we need to determine the number of atoms associated with each unit cell. Depending on an atom's location, it may be considered to be shared with adjacent unit cells—that is, only some fraction of the atom is assigned to a specific cell.
- For example, for cubic unit cells, an atom completely within the interior “belongs” to that unit cell, one at a cell face is shared with one other cell, and an atom residing at a corner is shared among eight.
- The number of atoms per unit cell, N , can be computed using the following formula:
- $N = N_i + N_f/2 + N_c/8$
- where
 - N_i the number of interior atoms
 - N_f the number of face atoms
 - N_c the number of corner atoms

- For the FCC crystal structure, there are eight corner atoms ($N_c = 8$), six face atoms ($N_f = 6$), and no interior atoms ($N_i = 0$). Thus, from Equation ,
- $N = 0 + 6/2 + 8/8 = 4$
- or a total of four whole atoms may be assigned to a given unit cell. This is depicted in Figure , where only sphere portions are represented within the confines of the cube.
- The cell is composed of the volume of the cube that is generated from the centers of the corner atoms, as shown in the figure.
- Corner and face positions are really equivalent—that is, translation of the cube corner from an original corner atom to the center of a face atom will not alter the cell structure.

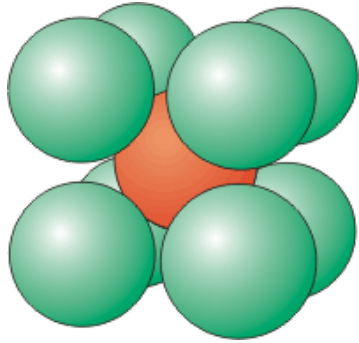
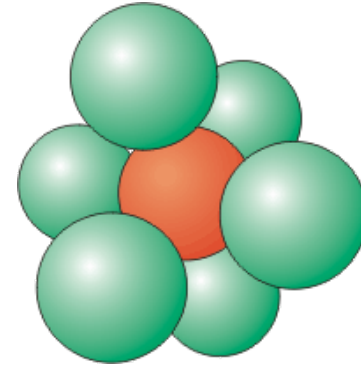


COORDINATION NUMBER

- The *coordination number* (CN) is the number of nearest neighbors of a given particle in the crystal lattice.
- It determines the nature of the bonding in a crystal.
- The most common coordination numbers are 4, 6, 8, and 12. respectively as shown below

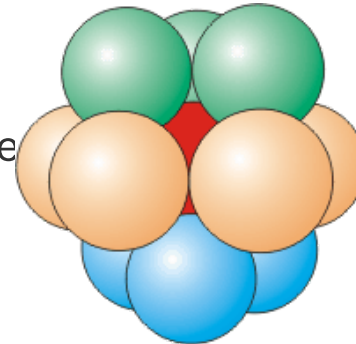


An atom on the corner of a simple cubic unit cell has a coordination number of six.



The atom in the center of a body-centered unit cell has a coordination number of eight.

Each atom in a face-centered cubic unit cell has a coordination number of 12. The 12 particles, which make up the *coordination sphere* of the red particle lie in three planes.

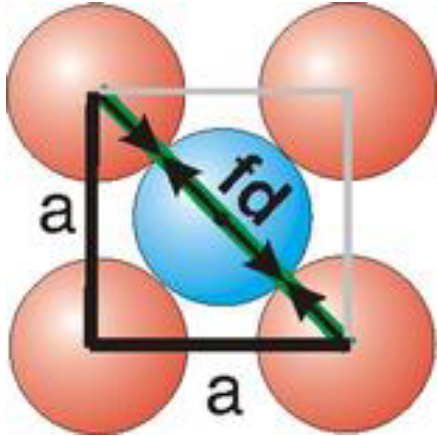
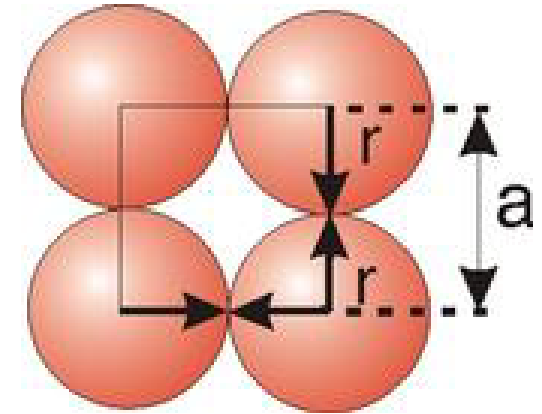


Atomic Radii

- Atoms are not hard spheres with distinct boundaries, so their sizes are not determined directly. However, the positions of the atoms in a solid can be determined by x-ray diffraction, and the sizes of the atoms are inferred from those distances.
- In this method, the radius of an atom is determined from the unit cell edge length (a), which is determined from the location of the atoms, and the assumption that the atoms touch as shown in Figures (next Slide).
- The radius obtained from the structure of a metallic solid is referred to as either the *metallic radius* or the *atomic radius*

Atomic Radii

$2r = a$ for simple cubic structure
Here r = atomic radius ; a = cube length

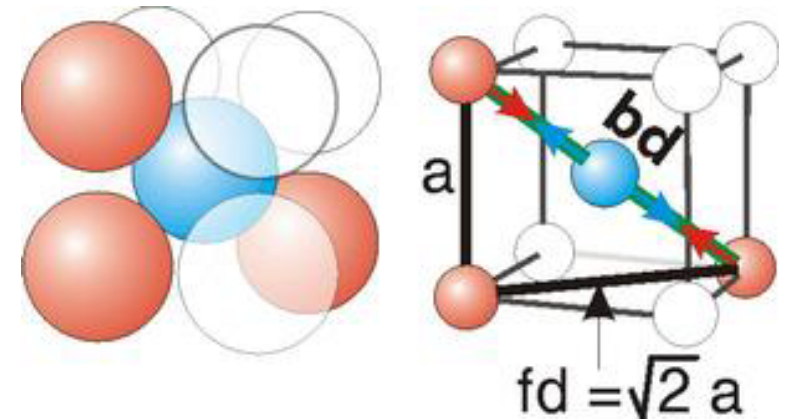


$$fd = 4r \rightarrow (fd)^2 = (a)^2 + (a)^2 \rightarrow (4r)^2 = 2(a)^2 \rightarrow r = \sqrt{2}a/4$$

For FCC structure

$$d = \sqrt{2}a \quad (bd)^2 = (d)^2 + (a)^2 \quad (4r)^2 = 3(a)^2$$

$r = \sqrt{3}a/4$ For BCC structure



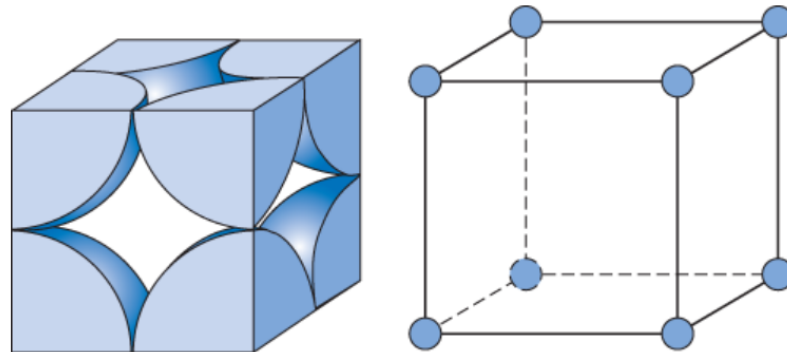
Determination of FCC Unit Cell Volume

• *Solution*

- In the FCC unit cell illustrated, the atoms touch one another across a face-diagonal, the length of which is $4r$. Because the unit cell is a cube, its volume is a^3 , where a is the cell edge length.
- From the right triangle on the face,
 - $a^2 + a^2 = 4r^2$
 - or, solving for a ,
 - $a = 2r \sqrt{2}$
- The FCC unit cell volume VC may be computed from
 - $VC = a^3 = (2r \sqrt{2})^3 = 16 r^3 \sqrt{2}$
 - Similarly find out for SC BCC & HCP

Atomic packing factor (APF).

- The APF is the sum of the sphere volumes of all atoms within a unit cell (assuming the atomic hard-sphere model) divided by the unit cell volume
- $$\text{APF} = \frac{\text{volume of atoms in a unit cell}}{\text{total unit cell volume}}$$
- Simple Cubic Structure:-In this structure, atoms touch each other along the lattice, as shown in Fig. $a = 2r$ or $r = a/2$

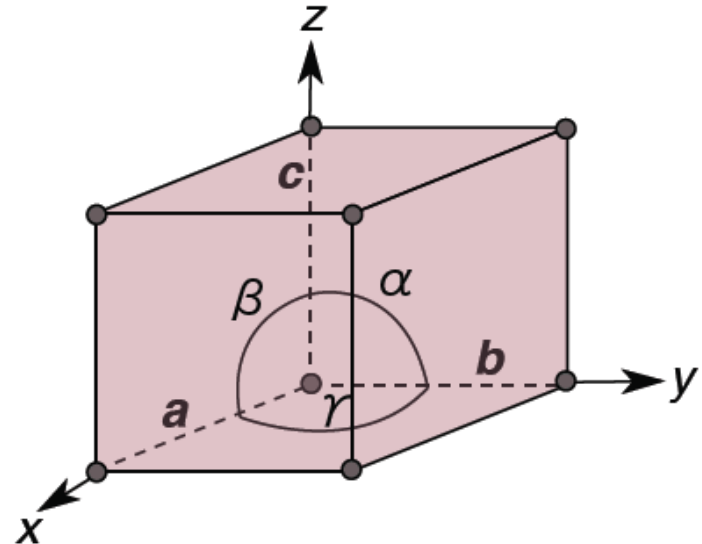


Computation of the APF for FCC

- $APF = \frac{\text{volume of atoms in a unit cell}}{\text{total unit cell volume}} = \frac{V_s}{V_c}$
- Both the total atom and unit cell volumes may be calculated in terms of the atomic radius r .
- The volume for a sphere $= \frac{4}{3}\pi r^3$
- and because there are four atoms per FCC unit cell, the total
- FCC atom (or sphere) volume is $V_s = (4) \times \frac{4}{3}\pi r^3 = \frac{16}{3}\pi r^3$
- From Example earlier, the total unit cell volume is $V_c = 16 r^3 \sqrt{2}$
- Therefore, the atomic packing factor is
- $APF = \frac{V_s}{V_c} = \frac{\frac{16}{3}\pi r^3}{16 r^3 \sqrt{2}} = 0.74$

Lattice geometry

- To define the geometry of the unit cell in 3 dimensions we choose a right-handed set of crystallographic axes, x , y , and z , which point along the edges of the unit cell. The origin of our coordinate system is at one of the lattice points.
- Lattice parameters
- The length of the unit cell along the x , y , and z direction are defined as a , b , and c . Alternatively, we can think of the sides of the unit cell in terms of vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$. The angles between the crystallographic axes are defined by:
 - α = the angle between $\mathbf{b}$ and $\mathbf{c}$
 - β = the angle between $\mathbf{a}$ and $\mathbf{c}$
 - γ = the angle between $\mathbf{a}$ and $\mathbf{b}$
- a , b , c , α , β , γ are collectively known as the **lattice parameters** (often also called ‘unit cell parameters’, or just ‘cell parameters’).



Crystal systems

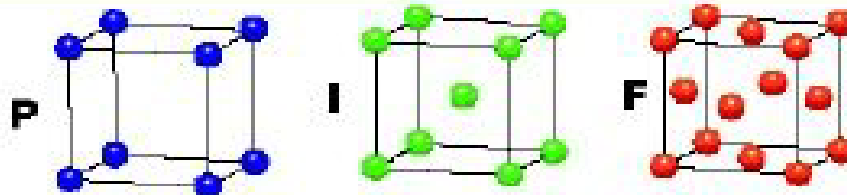
- The rotational symmetry of a crystal places constraints on the shape of the conventional unit cell we choose to describe the structure. On this basis we divide all structures into one of 7 crystal systems. For example, for crystals with 4 fold symmetry it will always be possible to choose a unit cell that has a square base with $a = b$ and $\gamma = 90^\circ$:
- There are 14 unique combinations of the 7 crystal systems with the possible types of primitive and non-primitive lattices. These are referred to as the 14 Bravais lattices.

Seven Crystal Structures

CUBIC

$$a = b = c$$

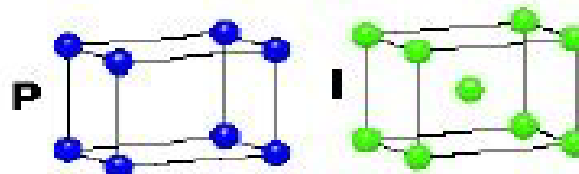
$$\alpha = \beta = \gamma = 90^\circ$$



TETRAGONAL

$$a = b \neq c$$

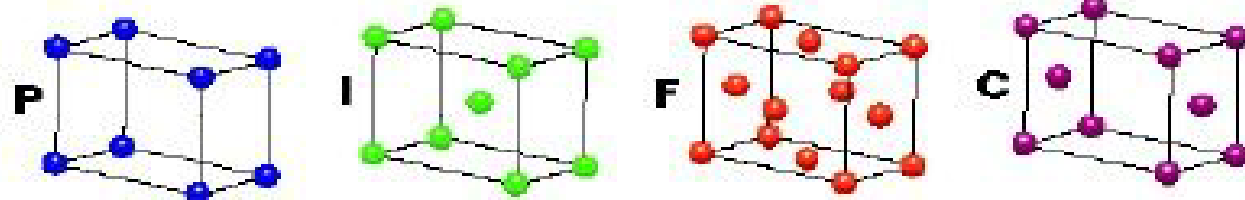
$$\alpha = \beta = \gamma = 90^\circ$$



ORTHORHOMBIC

$$a \neq b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$

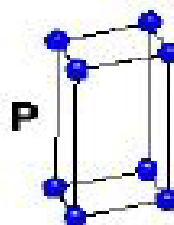


HEXAGONAL

$$a = b \neq c$$

$$\alpha = \beta = 90^\circ$$

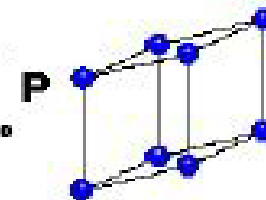
$$\gamma = 120^\circ$$



TRIGONAL

$$a = b = c$$

$$\alpha = \beta = \gamma \neq 90^\circ$$

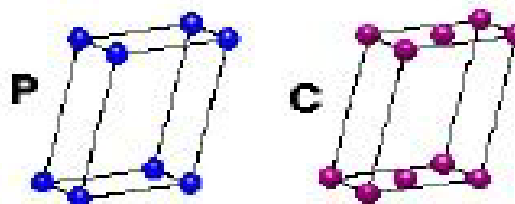


MONOCLINIC

$$a \neq b \neq c$$

$$\alpha = \gamma = 90^\circ$$

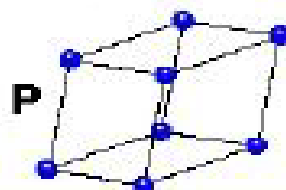
$$\beta \neq 120^\circ$$



TRICLINIC

$$a \neq b \neq c$$

$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$



4 Types of Unit Cell

P = Primitive

I = Body-Centred

F = Face-Centred

C = Side-Centred

+

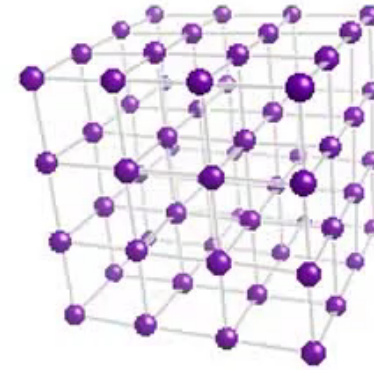
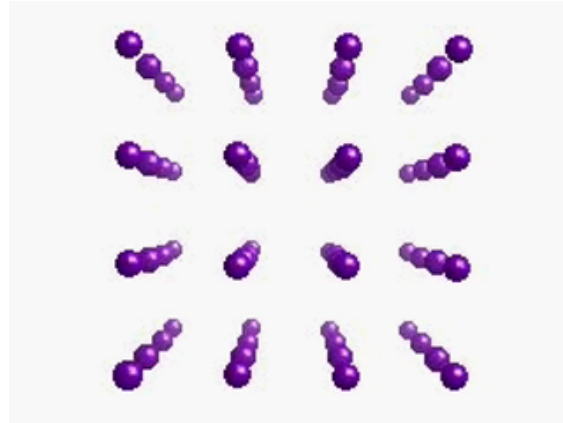
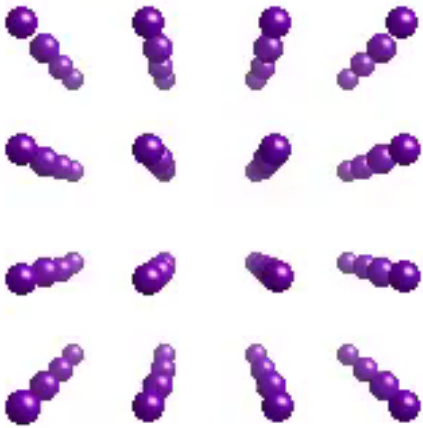
7 Crystal Classes

→ 14 Bravais Lattices

Miller Indices

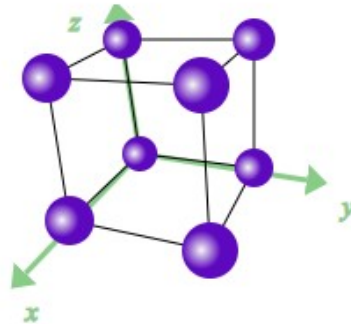
- Miller Indices are a method of describing the orientation of a plane or set of planes within a lattice in relation to the unit cell. They were developed by [William Hallows Miller](#).
- These indices are useful in understanding many phenomena in materials science, such as explaining the shapes of single crystals, the form of some materials' microstructure, the interpretation of X-ray diffraction patterns, and the movement of a dislocation , which may determine the mechanical properties of the material.

How to index a lattice plane



This is a 3D array of atoms in a primitive cubic lattice.

It can be represented as periodic unit cells.



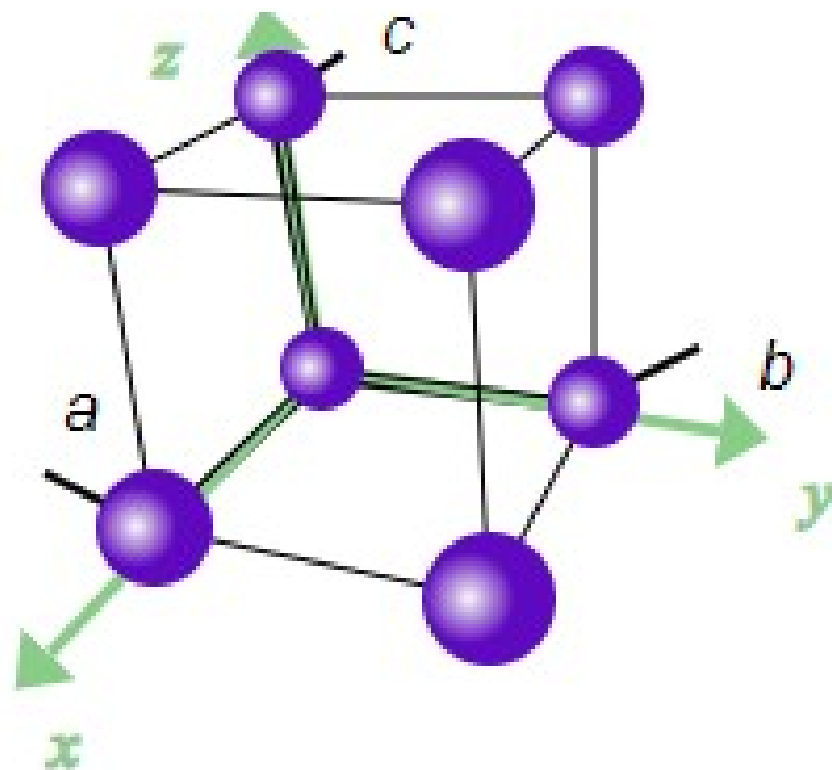
Now consider a single unit cell.
We can define a set of axes, x , y and z , that describe the edges of this unit cell.

The lengths of the edges are defined by distances, known as the lattice parameters:

a on the x -axis,

b on the y -axis,

c on the z -axis.



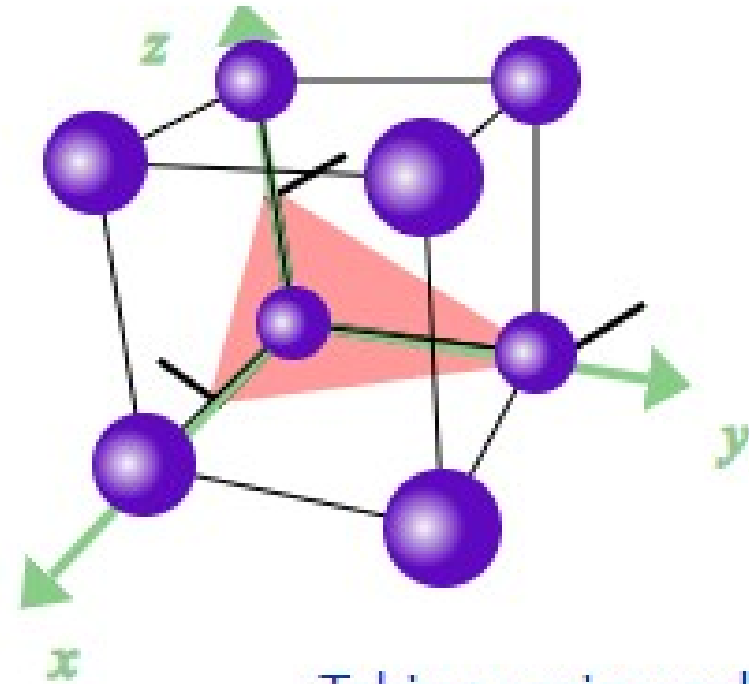
Suppose we have a plane within this unit cell.

This plane intercepts:

the x-axis at $a/2$,

the y-axis at $b/1$,

the z-axis at $c/2$,



Taking reciprocals of these intercepts gives the Miller indices:

(212)

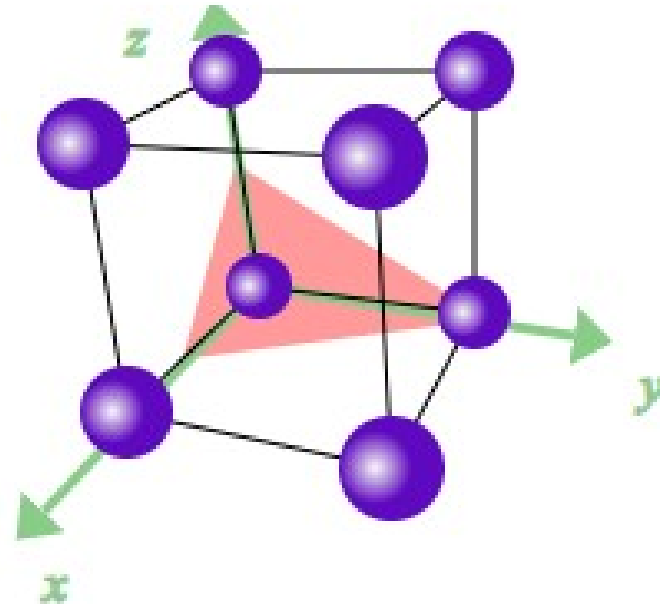


If the intercepts are:

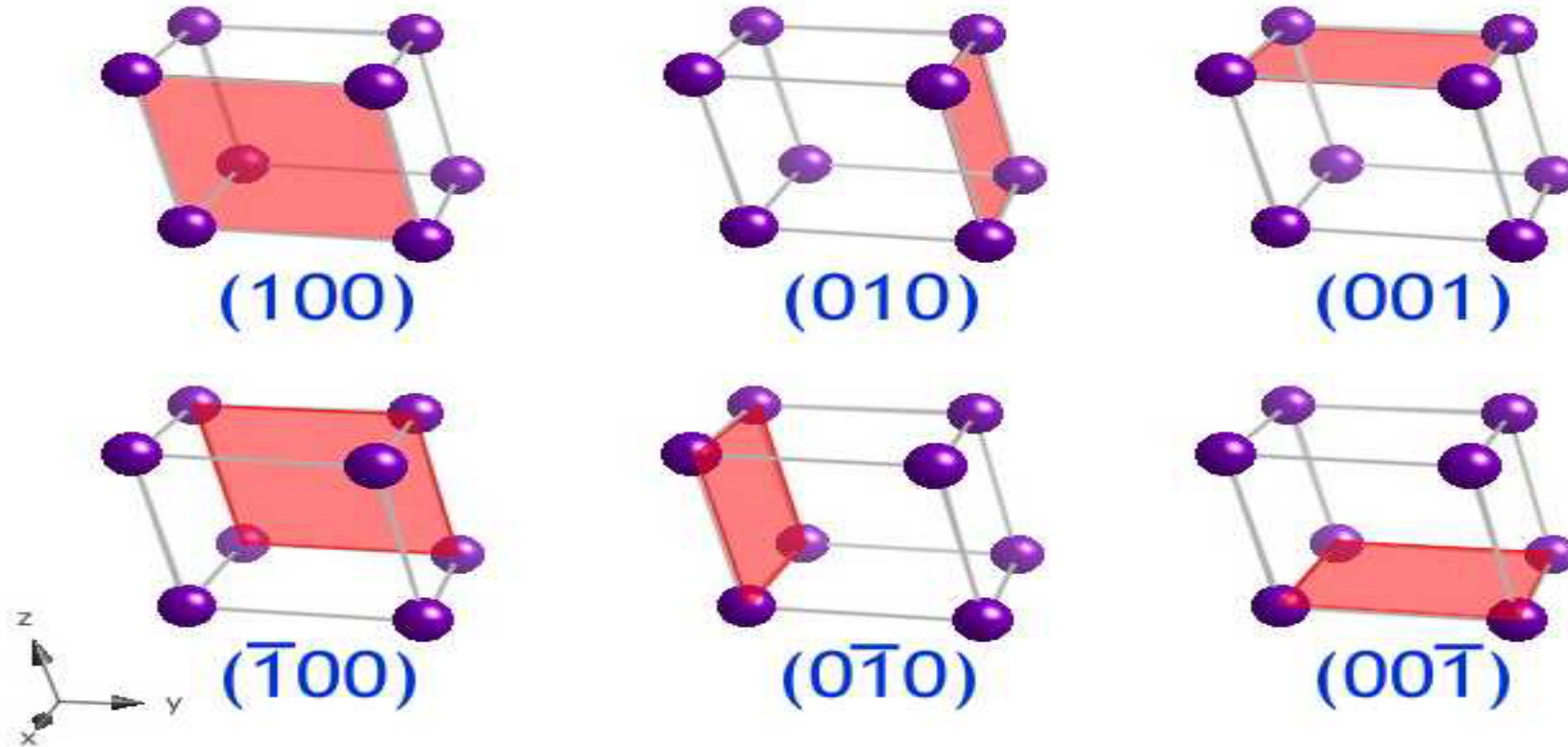
$1/h$ on the x-axis,
 $1/k$ on the y-axis,
 $1/l$ on the z-axis.

The Miller indices are (hkl)

Or, if the resulting indices contain fractions; multiply throughout to give integer values of h, k , and l .

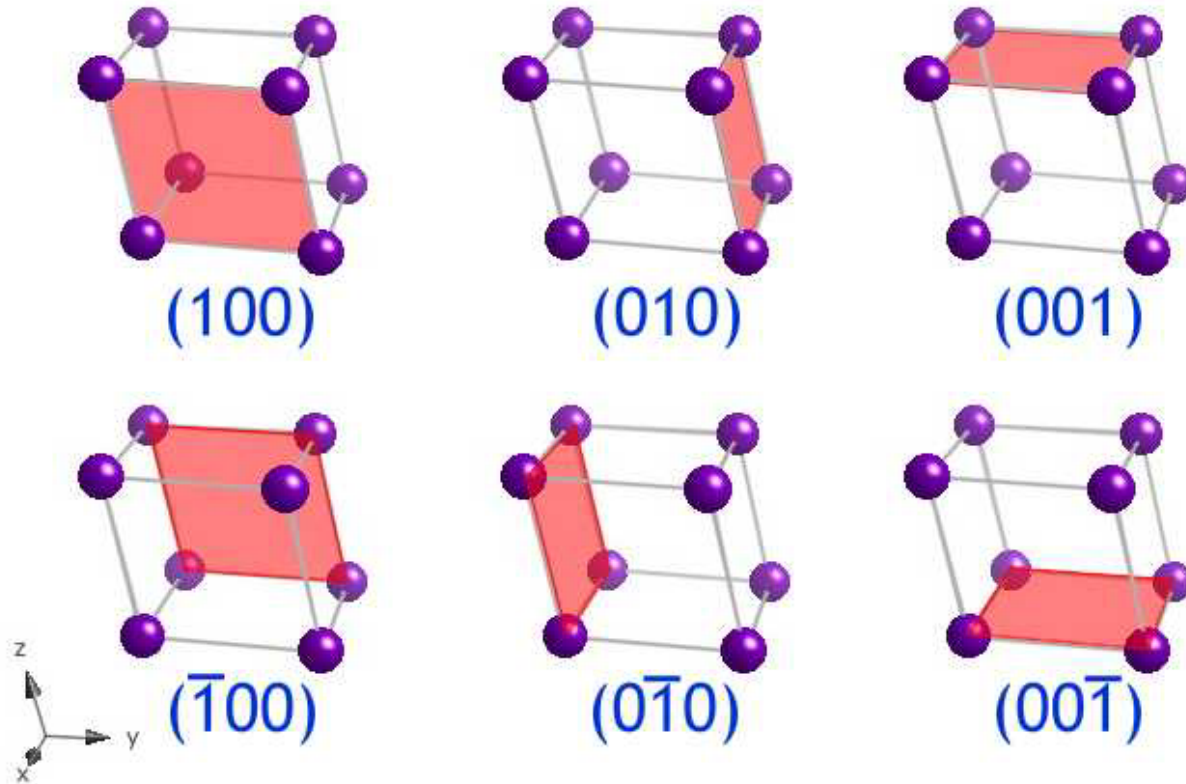


Examples of lattice planes



The (100) , (010) , (001) , $(\bar{1}00)$, $(0\bar{1}0)$ and $(00\bar{1})$ planes form the faces of the unit cell. Here, they are shown as the faces of a triclinic ($a \neq b \neq c$, $\alpha \neq \beta \neq \gamma$) unit cell. Although in this image, the (100) and $(\bar{1}00)$ planes are shown as the front and back of the unit cell, both indices refer to the same family of planes

Examples of lattice planes



The (101) , (110) , (011) , $(\bar{1}01)$, $(1\bar{1}0)$ and $(01\bar{1})$ planes form the sections through the diagonals of the unit cell, along with those planes whose indices are the negative of these. In the image the planes are shown in a different triclinic unit cell.

End of Lecture

Thanks

Queries if any >>>>>

Imperfection in Crystals

Dr. Dinesh Shringi

Content

- Imperfection in crystals.
- point defects – vacancies, Schottky
- Frenkel and line defects .
- dislocations, edge, screw.
- Burgers vector.

Introduction

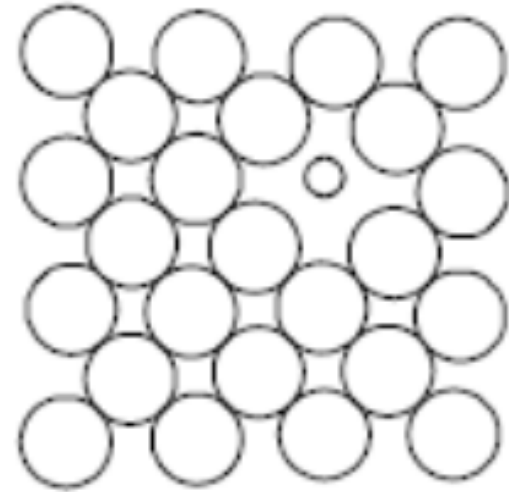
- Thus far it has been tacitly assumed that perfect order exists throughout crystalline materials on an atomic scale. However, such an idealized solid does not exist; all contain large numbers of various defects or **imperfections**. As a matter of fact, many of the properties of materials are profoundly sensitive to deviations from crystalline perfection.
- The influence is not always adverse, and often specific characteristics are deliberately fashioned by the introduction of controlled amounts or numbers of particular defects, as detailed in succeeding chapters.
- A *crystalline defect* refers to a lattice irregularity having one or more of its dimensions on the order of an atomic diameter. Classification of crystalline imperfections is frequently made according to the geometry or dimensionality of the defect.
- Several different imperfections are discussed in this Session, including **point defects** (those associated with one or two atomic positions); linear (or one-dimensional) defects; and interfacial defects, or boundaries, which are two-dimensional. Impurities in solids are also discussed, because impurity atoms may exist as point defects. Finally, techniques for the microscopic examination of defects and the structure of materials are briefly described.

Point Defects

- The point imperfections, which are lattice errors at isolated lattice points, take place due to imperfect packing of atoms during crystallisation.
- The point imperfections also take place due to vibrations of atoms at high temperatures.
- Point imperfections are completely local in effect, e.g. a vacant lattice site.
- Point defects are always present in crystals and their presence results in a decrease in the free energy.

VACANCIES

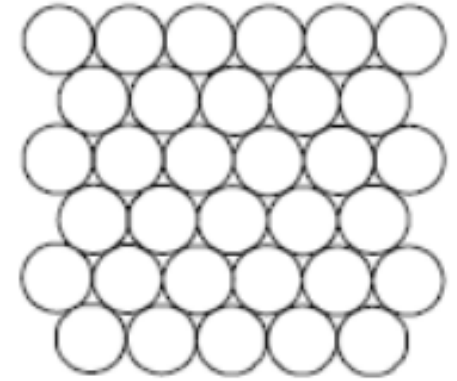
- The simplest of the point defects is a **vacancy**, or vacant lattice site, one normally occupied but from which an atom is missing .
- All crystalline solids contain vacancies, and, in fact, it is not possible to create such a material that is free of these defects.
- The necessity of the existence of vacancies is explained using principles of thermodynamics; in essence, the presence of vacancies increases the entropy (i.e., the randomness) of the crystal.



(a) Vacancy defect

self-interstitial

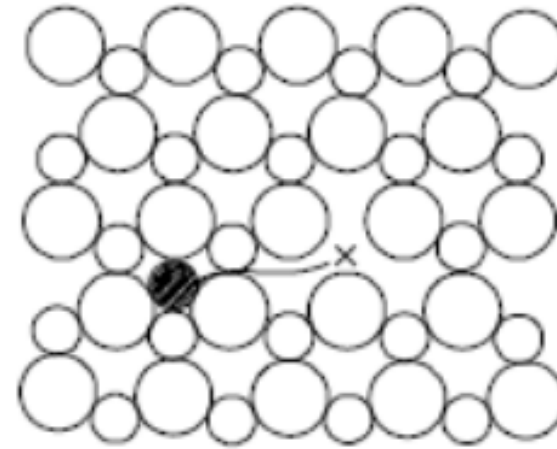
- In a closed packed structure of atoms in a crystal if the atomic packing factor is low, an extra atom may be lodged within the crystal structure. This is known as *interstitial position*, i.e. *voids*.
- An extra atom can enter the interstitial space or void between the regularly positioned atoms only when it is substantially smaller than the parent atoms, otherwise it will produce atomic distortion. The defect caused is known as interstitial defect.
- In close packed structures, e.g. FCC and HCP, the largest size of an atom that can fit in the interstitial void or space have a radius about 22.5% of the radii of parent atoms.
- Interstitialcies may also be single interstitial, di-interstitials, and tri-interstitials. We must note that vacancy and interstitialcy are inverse phenomena.



(b) Interstitial defect

Frenkel Defect

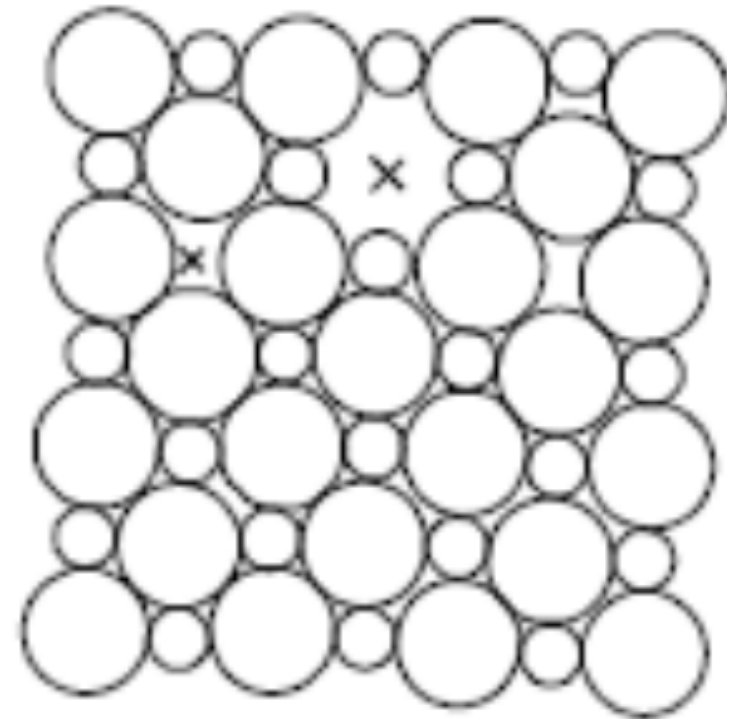
- Whenever a missing atom, which is responsible for vacancy occupies an interstitial site (responsible for interstitial defect) as shown in Fig. (c), the defect caused is known as Frenkel defect.
- Obviously, Frenkel defect is a combination of vacancy and interstitial defects. These defects are less in number because energy is required to force an ion into new position.
- This type of imperfection is more common in ionic crystals, because the positive ions, being smaller in size, get lodged easily in the interstitial positions.



(c) Frenkel defect

Schottky Defect:

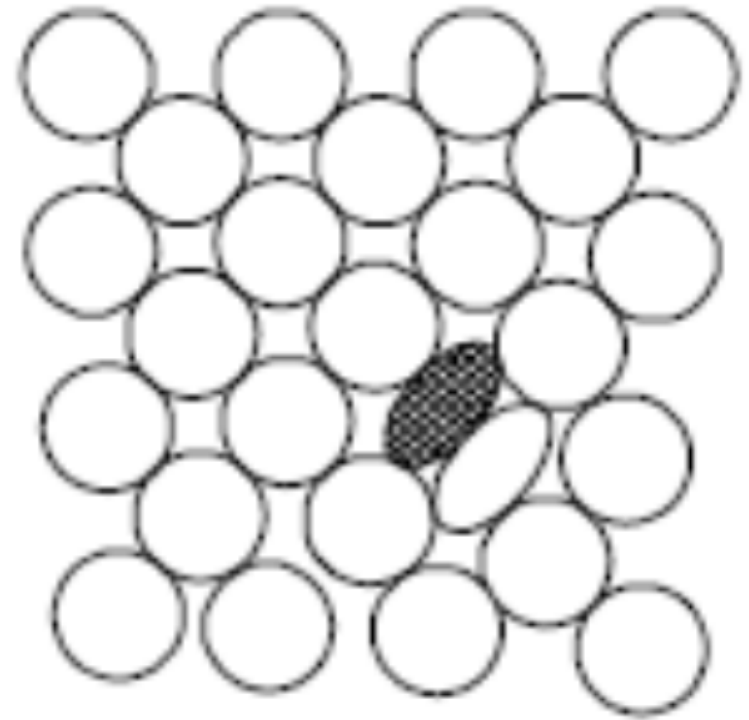
- These imperfections are similar to vacancies. This defect is caused, whenever a pair of positive and negative ions is missing from a crystal (e).
- This type of imperfection maintains a charge neutrality. Closed-packed structures have fewer interstitialcies and Frenkel defects than vacancies and Schottky defects, as additional energy is required to force the atoms in their new positions.



(e) Schottky defect

Substitutional Defect:

- Whenever a foreign atom replaces the parent atom of the lattice and thus occupies the position of parent atom ((d)], the defect caused is called substitutional defect.
- In this type of defect, the atom which replaces the parent atom may be of same size or slightly smaller or greater than that of parent atom.



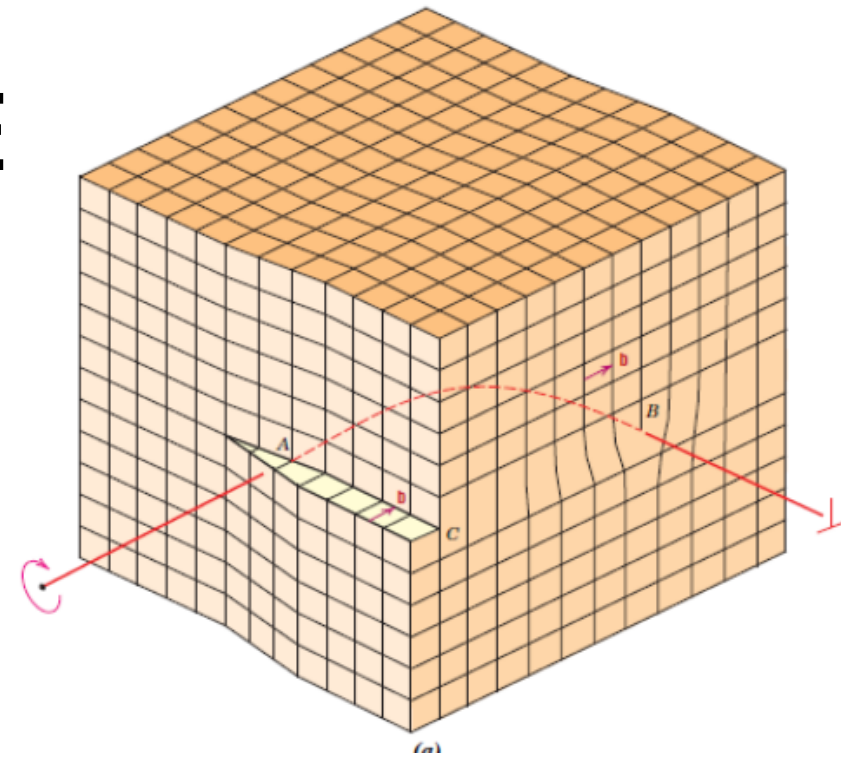
(d) Substitutional defect

Line Deflections

- Line imperfections are called dislocations. A linear disturbance, i. e. one dimensional imperfections in the geometrical sense of the atomic arrangement, which can very easily occur on the slip plane through the crystal, is known as dislocation.
- The most important kinds of linear defects are *edge* and *screw dislocation*.
- Both these types are formed in the process of their deformation. Both these defects are the most striking imperfections and are responsible for the useful property of ductility in metals, ceramics and crystalline polymers.

DISLOCATIONS-LINEAR DEFECTS

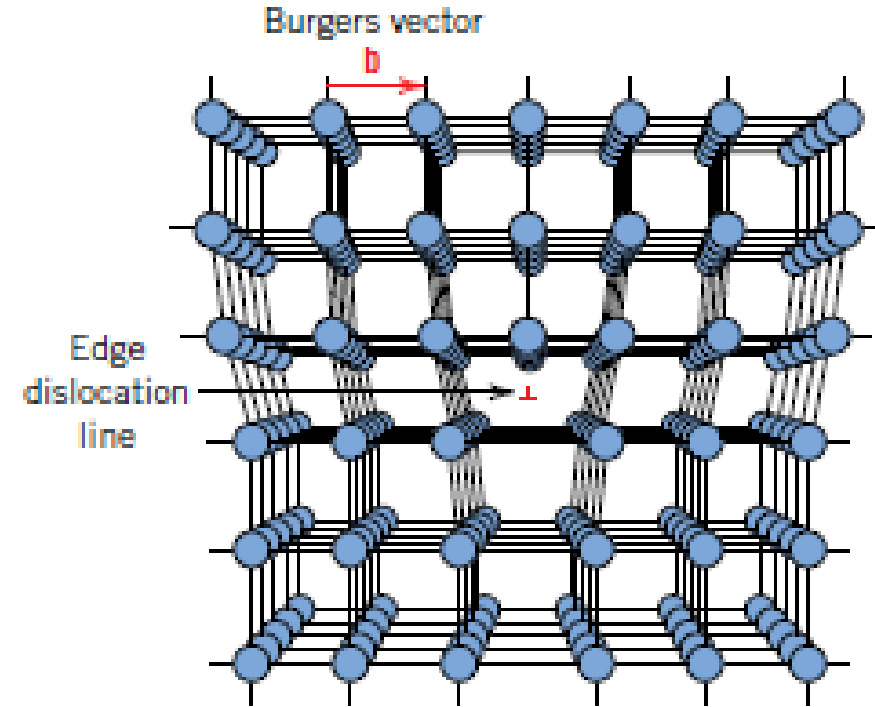
- A *dislocation* is a linear or one-dimensional defect around which some of the atoms are misaligned.
- They are of two types
 - Edge Dislocation
 - Screw Dislocation
- Most dislocations found in crystalline materials are probably neither pure edge nor pure screw but exhibit components of both types; these are termed **mixed dislocations**.



At point A, the dislocation is pure screw, while at point B, it is pure edge. For regions in between where there is curvature in the dislocation line, the character is mixed edge and screw.

Edge dislocation

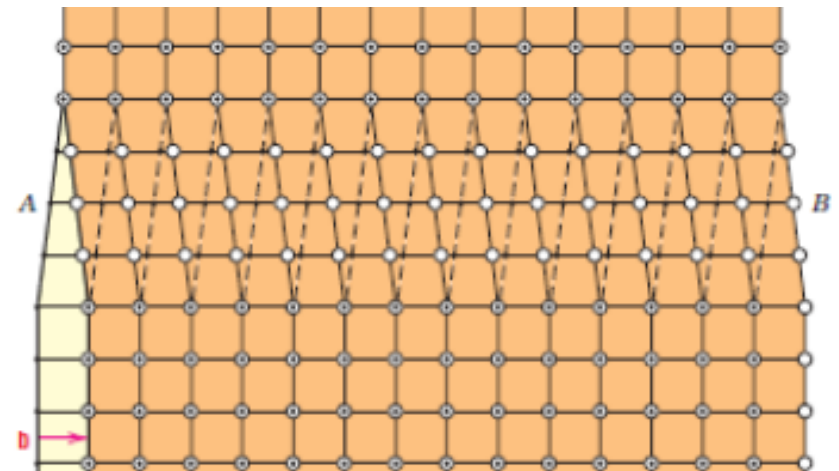
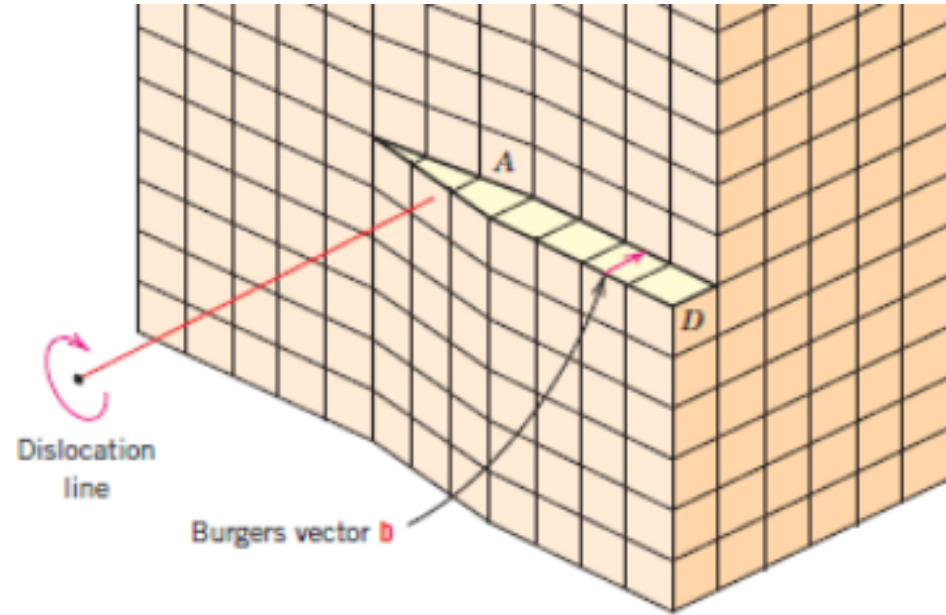
- an extra portion of a plane of atoms, or half-plane, the edge of which terminates within the crystal. This is termed an **edge dislocation**;
- it is a linear defect that centers on the line that is defined along the end of the extra half-plane of atoms. This is sometimes termed the **dislocation line**, is perpendicular to the plane of the page. Within the region around the dislocation line there is some localized lattice distortion.
- The atoms above the dislocation line are squeezed together, and those below are pulled apart; this is reflected in the slight curvature for the vertical planes of atoms as they bend around this extra half-plane.
- The magnitude of this distortion decreases with distance away from the dislocation line; at positions far removed, the crystal lattice is virtually perfect. An edge dislocation may also be formed by an extra half-plane of atoms that is included in the bottom portion of the crystal



The atom positions around an edge dislocation; extra half-plane of atoms shown in perspective.

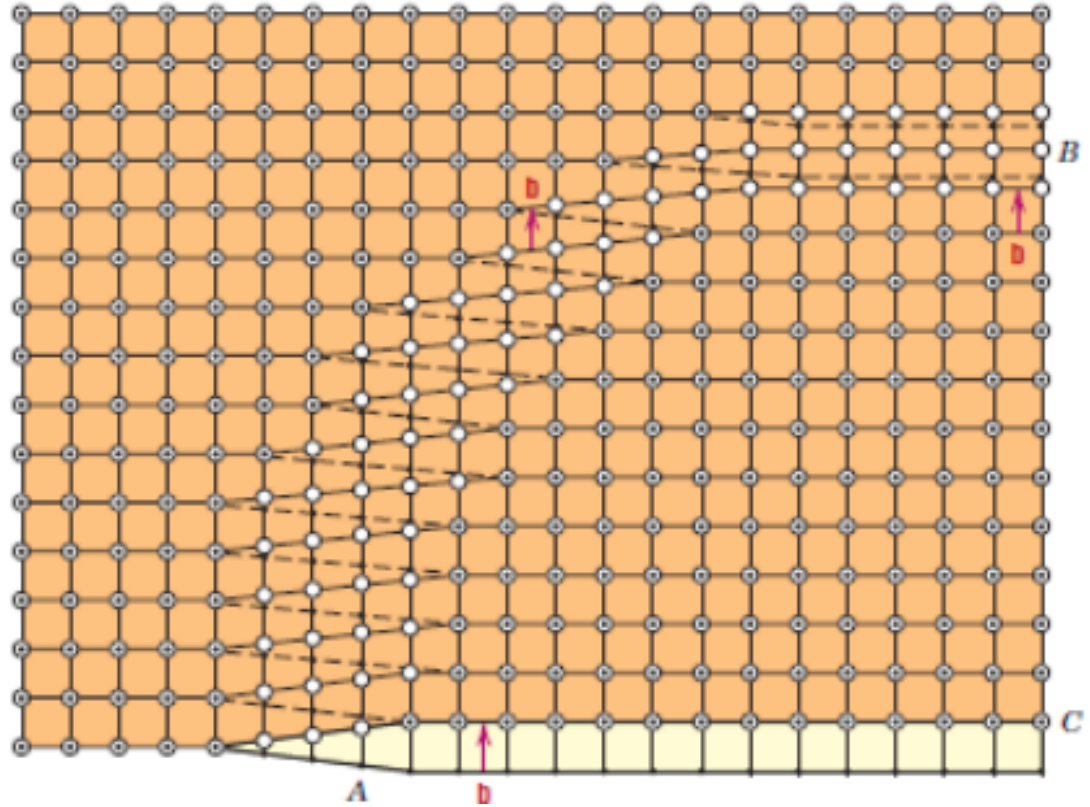
Screw dislocation

- Another type of dislocation, called a **screw dislocation**, may be thought of as being formed by a shear stress that is applied to produce the distortion
- the upper front region of the crystal is shifted one atomic distance to the right relative to the bottom portion.
- The atomic distortion associated with a screw dislocation is also linear and along a dislocation line.
- The screw dislocation derives its name from the spiral or helical path or ramp that is traced around the dislocation line by the atomic planes of atoms.



Mixed dislocations

- Most dislocations found in crystalline materials are probably neither pure edge nor pure screw but exhibit components of both types; these are termed **mixed dislocations**.
- All dislocation types are in the lattice distortion that is produced away from the two faces is mixed, having varying degrees of screw and edge character.

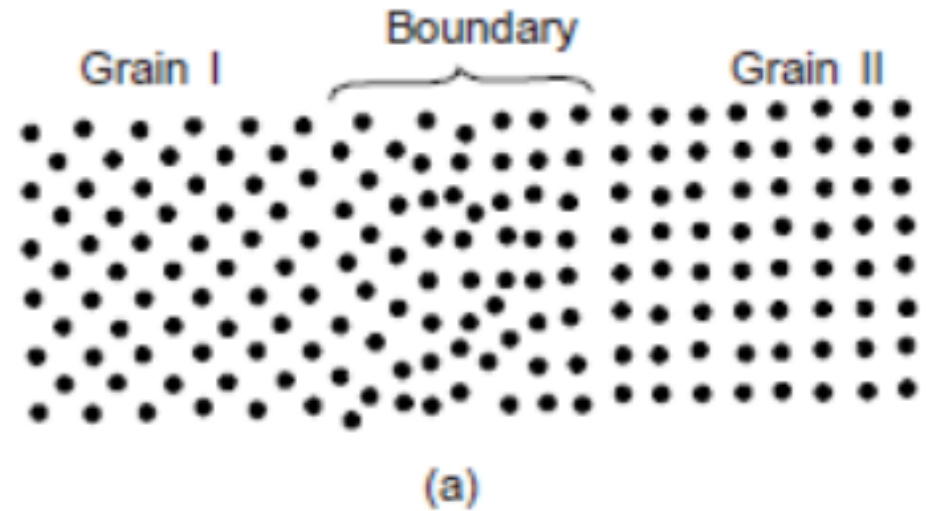


Surface and grain Boundary

- Surface and grain boundary imperfections of a structural nature arise from a change in the stacking of atomic planes on or across a boundary and are two-dimensional. The change may be one of the orientations or of the stacking sequence of the planes.
- The most important kinds of surface defects are high-angle and low angle boundaries, stacking faults, and twin boundaries. Basically surface imperfections are of two types: *external* and *internal*.
- The *external surface* of a material is an *imperfection* itself because the atomic bonds do not extend beyond it. The surface atoms have neighbours on one side only while atoms inside the crystal have neighbours on either side of them (Fig. 3.53). Since surface atoms are not entirely surrounded by others, they possess higher energy than that of internal atoms. For most metals, the energy of surface atom is of the order 1 J/m^2 .
- The *internal surface imperfections* are manifested by such defects as grain boundaries, tilt boundaries, twin boundaries, and stacking faults.

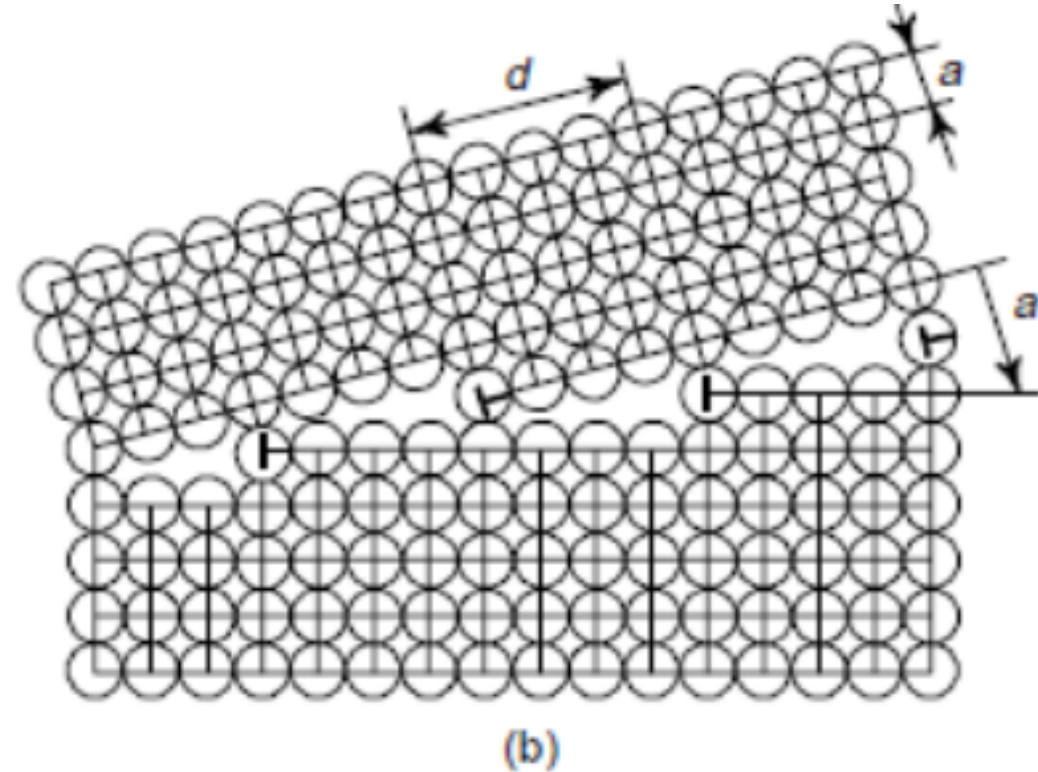
Grain Boundaries

- Engineering materials may be either polycrystalline or single crystal type. A polycrystalline alloy contains an enormous quantity of fine grains. Grain boundary imperfections are those surface imperfections which separate crystals or grains of different orientation in a polycrystalline aggregation during nucleation or crystallization. The shape of a grain is usually influenced by the presence of surrounding grains.
- The lattices of adjacent grains are oriented at random and differently (Fig.) and a boundary between any two grains is essentially a transition layer of thickness of 1–5 nm. This layer may have a disordered arrangement of atoms, dislocation of clusters, and an elevated concentration of impurities.



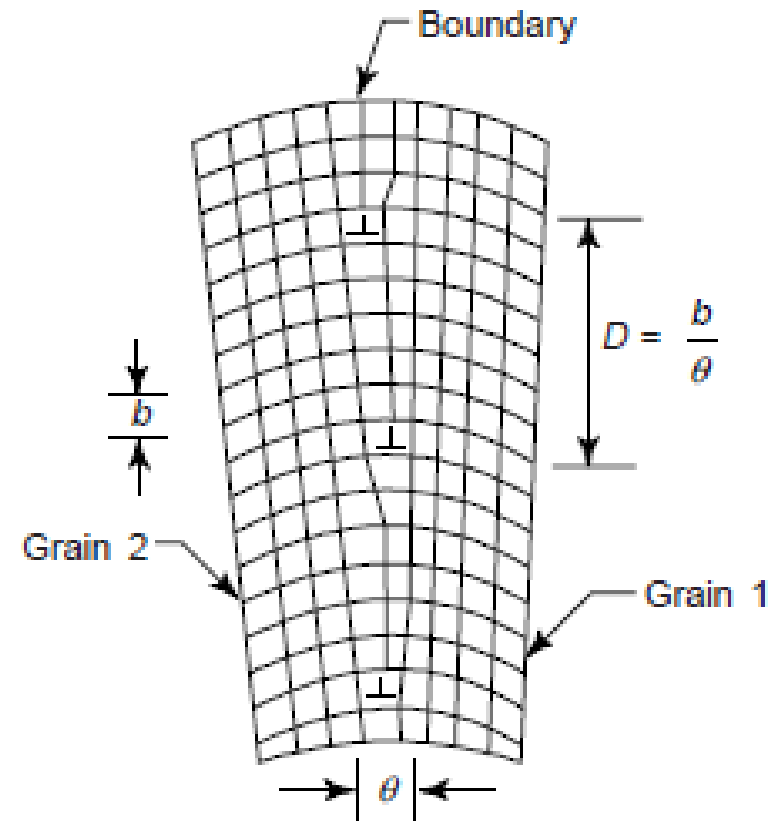
Grain Boundaries...

- In grain boundaries the atomic packing is imperfect and between two adjacent grains, there is a transition zone that is not aligned with either grain (Fig.). As a result, we note that boundary atoms in such cases cannot have a perfect complement of surrounding atoms.
- The mismatch with orientation of the adjacent grain produces a less efficient packing of atoms along the boundary. Thus, the atoms along the boundary have a higher energy than those within the grains. The higher energy of the boundary atoms is also important for the nucleation of polymorphic phase changes. The boundary between two crystals which have different crystalline arrangements or different compositions, is called an *interphase* boundary or commonly called an *interface*.



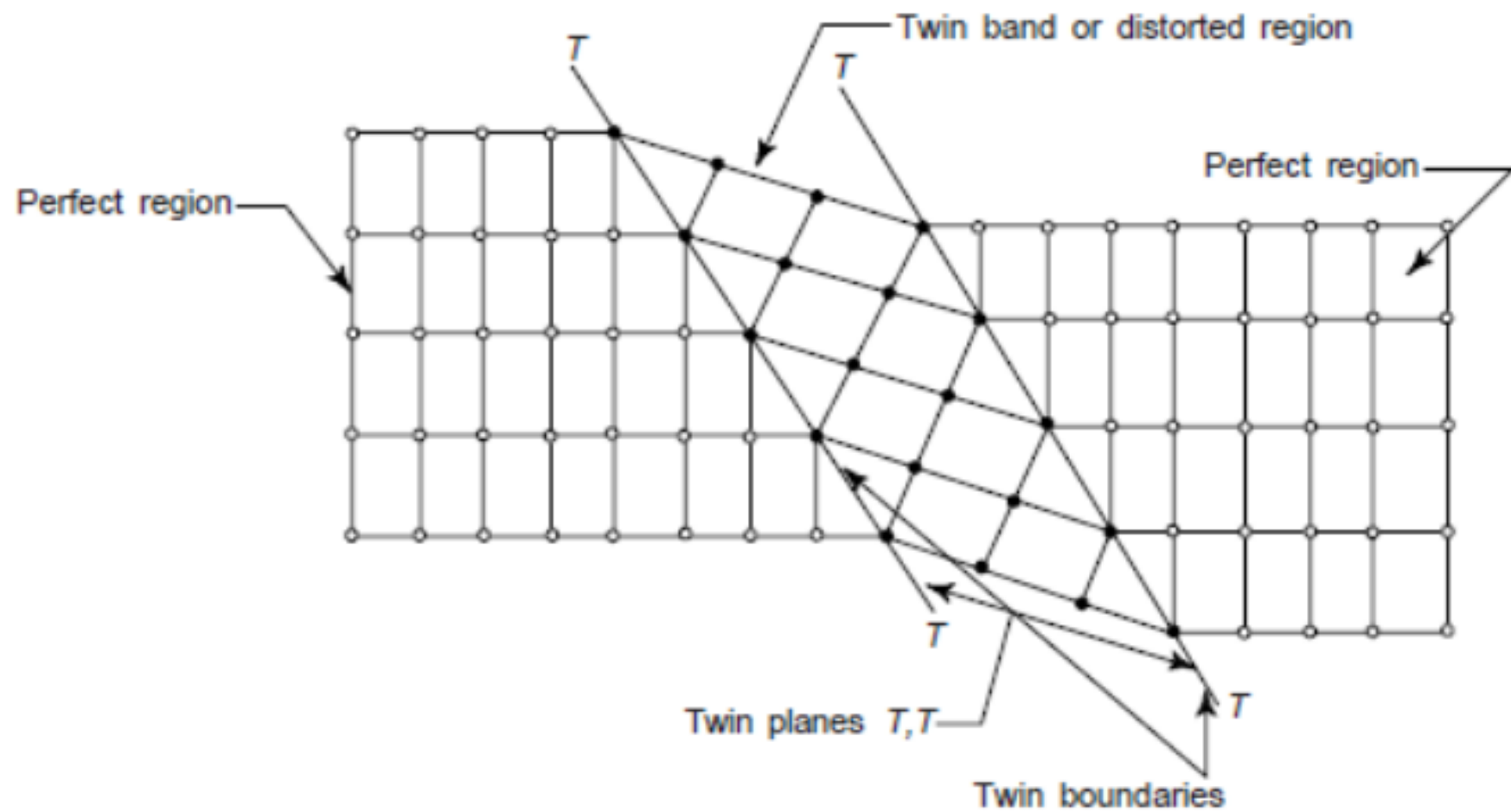
Tilt Boundaries

- This is another type of surface defect called low-angle boundary as the orientation difference between two neighbouring crystals is less than 10° . This is why the disruption in the boundary is not so drastic as in the high angle boundary.
- This type of boundary is associated with relatively little energy and is composed of edge dislocations lying one above the other. In general, one can describe low-angle boundaries by suitable arrays of dislocation.
- The angle or tilt, $\theta = b/D$ as in Fig. , where b is the magnitude of Burgers vector and D is the average vertical distance between dislocations.



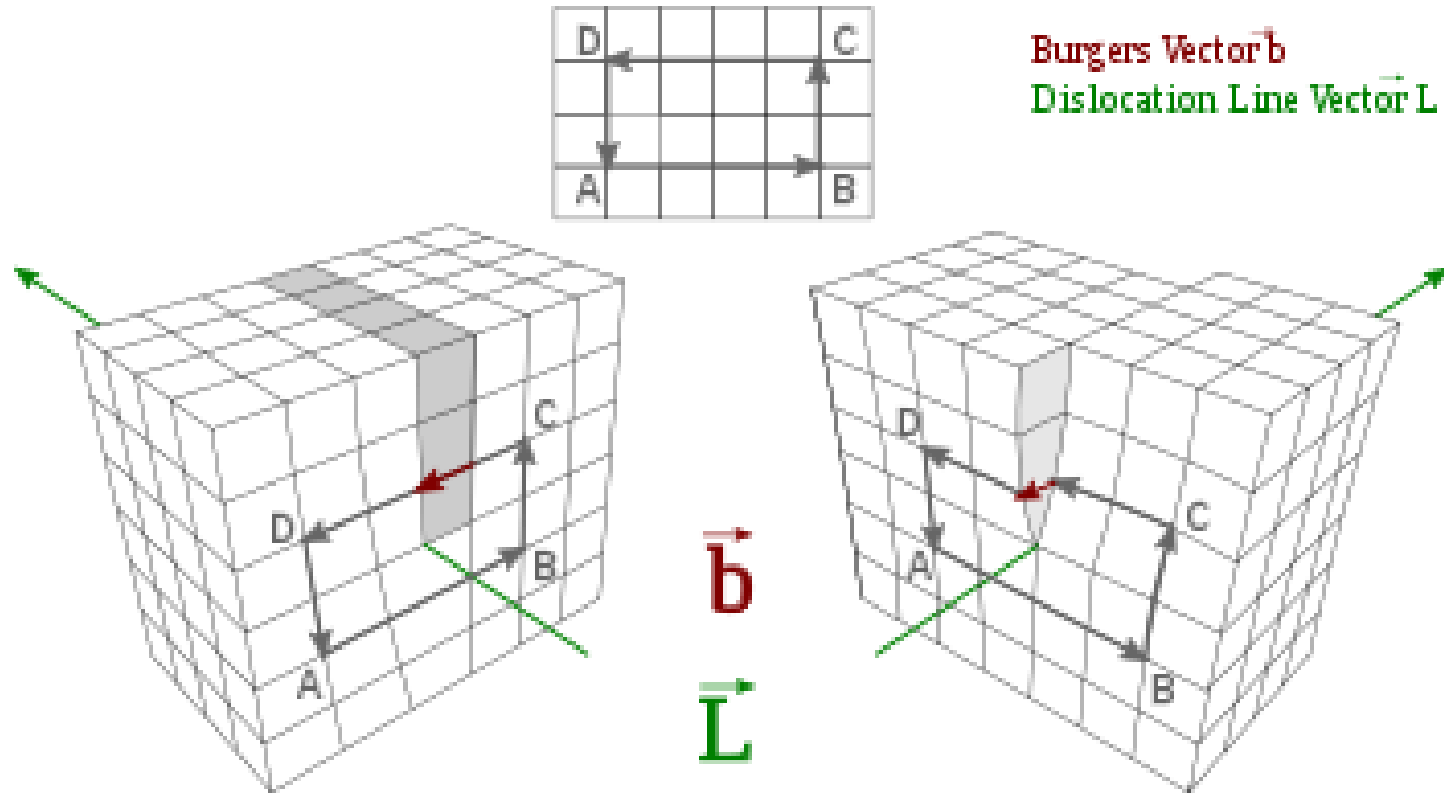
Twin Boundaries

- This is another planar surface imperfection. The atomic arrangement on one side of a twin boundary is a mirror reflection of the arrangement on the other side. Twinning may result during crystal growth or deformation of materials.
- Twin boundaries occur in pairs, such that the orientation change introduced by one boundary is restored by the other. The region between the pair of boundaries is termed as the *twinned region*. One can easily identify twin boundaries under an optical microscope.
- Twins which form during the process of recrystallization, i.e., in the process of mechanical working are known as mechanical twins, whereas twins formed as a result of annealing after plastic deformation are known as annealing twins (Fig.).



Burger's Vector

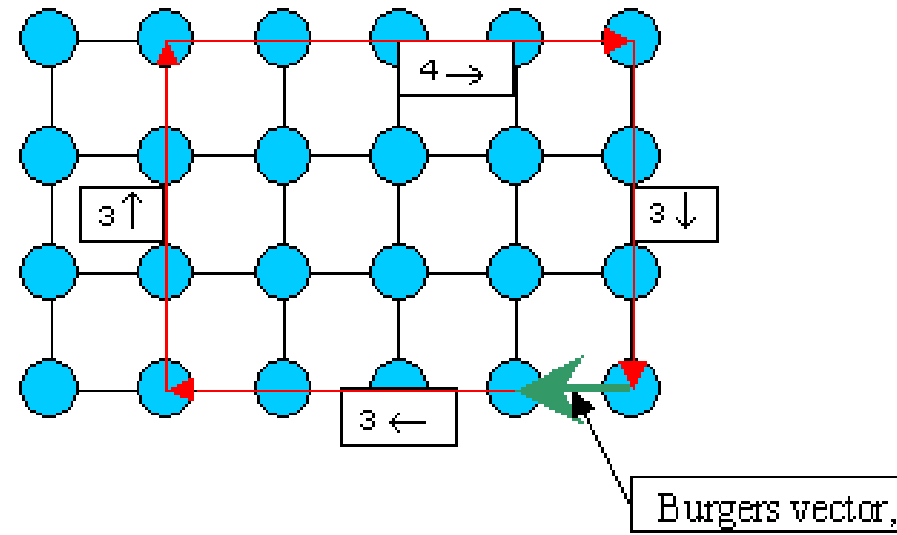
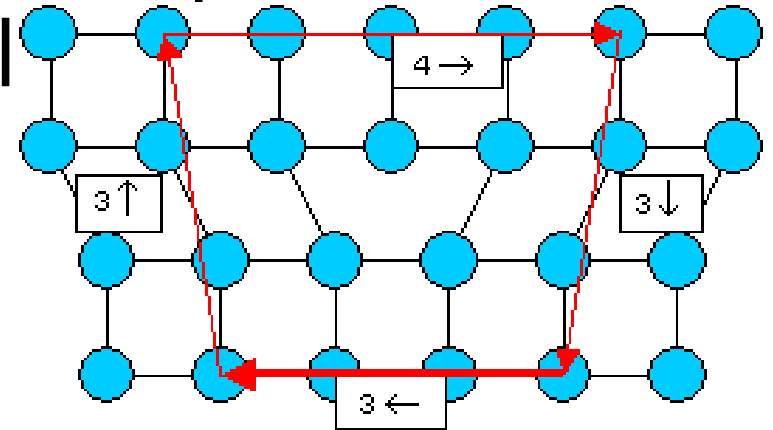
- The magnitude and direction of the lattice distortion associated with a dislocation are expressed in terms of a **Burgers vector**, denoted by **b**.
- Burgers vectors are indicated in Figures for edge and screw dislocations, respectively. Furthermore, the nature of a dislocation (i.e., edge, screw, or mixed) is defined by the relative orientations of dislocation line and Burgers vector.
- For an edge, they are perpendicular, whereas for a screw, they are parallel; they are neither perpendicular nor parallel for a mixed dislocation.



Burgers vector in an edge dislocation (left) and in a screw dislocation (right). The edge dislocation can be imagined as the introduction of a half plane (gray boxes) that does not fit the crystal symmetry. The screw dislocation can be imagined as cut and shear operation along a half plane

To determine the Burgers vector of a dislocation in a two-dimensional primitive square lattice, proceed as follows:

- Trace around the end of the dislocation plane to form a closed loop. Record the number of lattice vectors travelled along each side of the loop (shown here by the numbers in the boxes):
- In a perfect lattice, trace out the same path, moving the same number of lattice vectors along each direction as before. This loop will not be complete, and the closure failure is the Burgers vector:



- Thanks
- End of Lecture

Plastics Deformation of Metals and Alloys

Dr. Dinesh Shringi

Contents

- **Plastics Deformation of Metals and Alloys:**
- Internal stresses.
- role of dislocation.
- slip.
- Frank-Read source.
- Twinning.
- Strain hardening.

Why Deformation Studies

- With a knowledge of the nature of dislocations and the role they play in the plastic deformation process,
- we are able to understand the underlying mechanisms of the techniques that are used to strengthen and harden metals and their alloys;
- thus, it becomes possible to design and tailor the mechanical properties of materials—
- for example, the strength or toughness of a metal-matrix composite.
- Also, understanding the mechanisms by which polymers elastically and plastically deform allows one to alter and control their moduli of elasticity and strengths

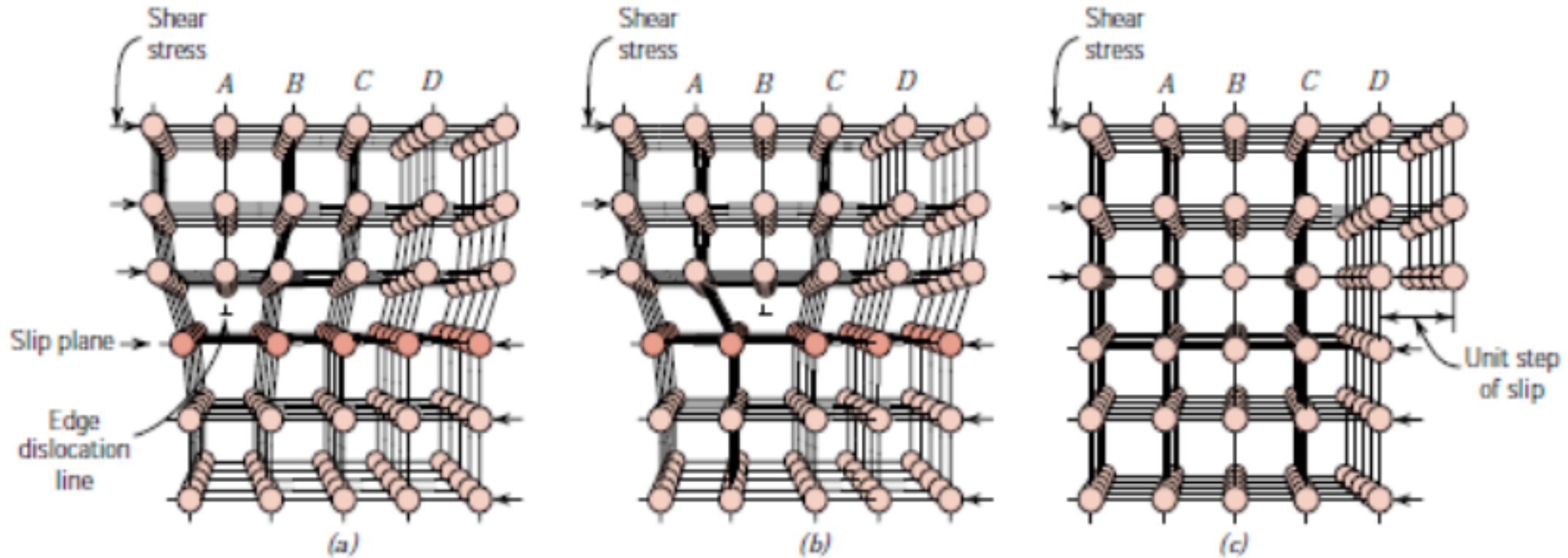
DEFORMATION MECHANISMS FOR METALS

- metallic materials may experience two kinds of deformation:
- elastic and plastic.
- Plastic deformation is permanent, and strength and hardness are measures of a material's resistance to this deformation.
- On a microscopic scale, plastic deformation corresponds to the net movement of large numbers of atoms in response to an applied stress.
- During this process, interatomic bonds must be ruptured and then reformed. Furthermore, plastic deformation most often involves the motion of dislocations, linear crystalline defects

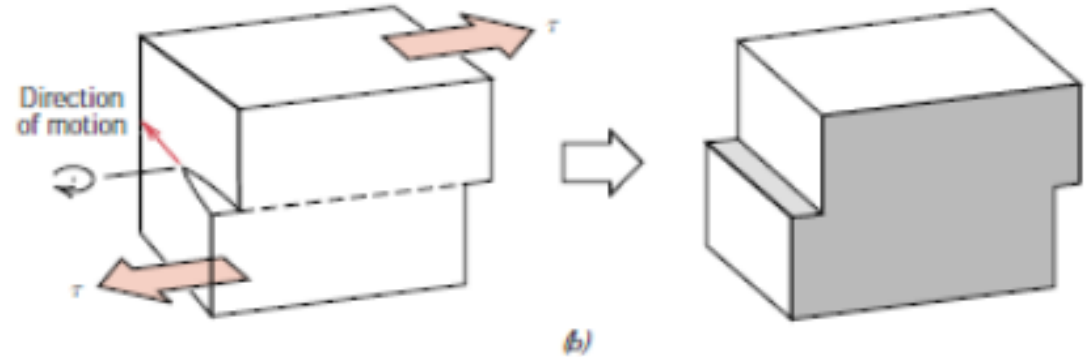
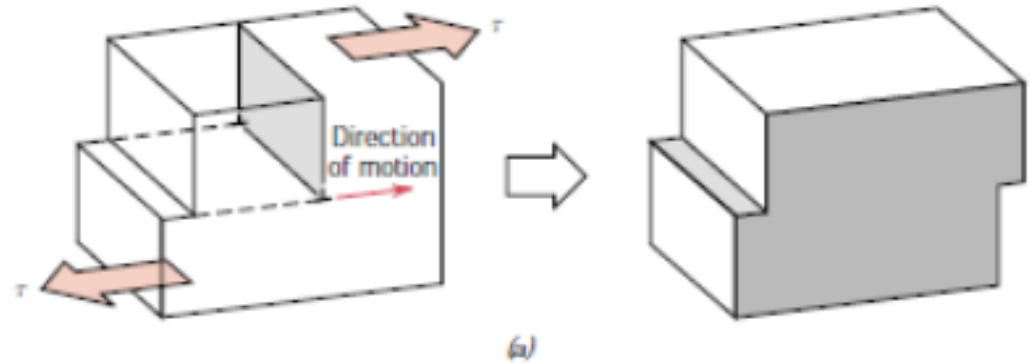
BASIC CONCEPTS OF DISLOCATIONS

- Edge and screw are the two fundamental dislocation types. In an edge dislocation, localized lattice distortion exists along the end of an extra half-plane of atoms, which also defines the dislocation line (Figure).
- A screw dislocation may be thought of as resulting from shear distortion; its dislocation line passes through the center of a spiral, atomic plane ramp (Figure).
- Many dislocations in crystalline materials have both edge and screw components; these are mixed dislocations (Figure)
-

- Atomic rearrangements that accompany the motion of an edge dislocation as it moves in response to an applied shear stress. (a) The extra halfplane of atoms is labeled A. (b) The dislocation moves one atomic distance to the right as A links up to the lower portion of plane B; in the process, the upper portion of B becomes the extra half-plane. (c) A step forms on the surface of the crystal as the extra half-plane exits.



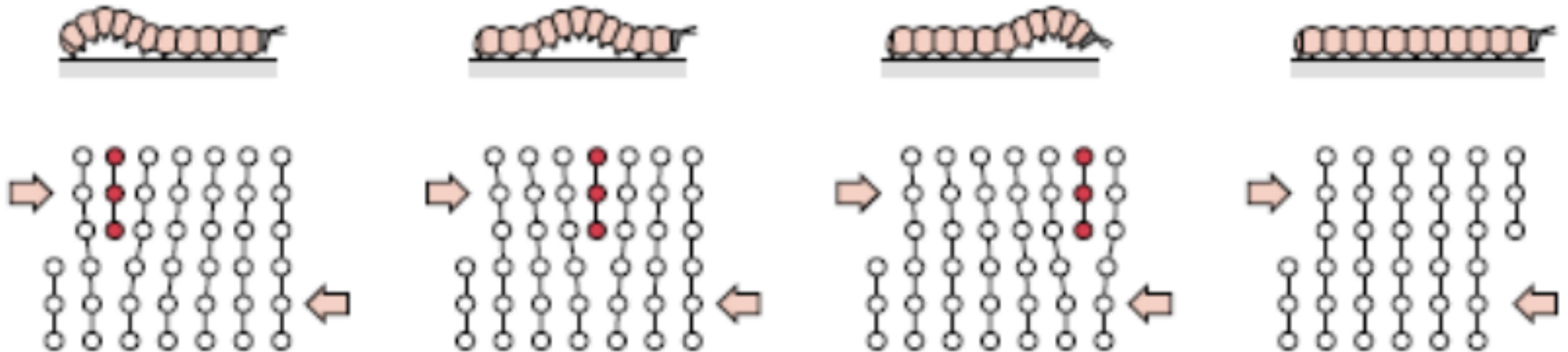
- The formation of a step on the surface of a crystal by the motion of (a) an edge dislocation and (b) a screw dislocation.
- Note that for an edge, the dislocation line moves in the direction of the applied shear stress ;
- for a screw, the dislocation line motion is perpendicular to the stress direction.



Slip

- The process by which plastic deformation is produced by dislocation motion is termed **slip**;
- the crystallographic plane along which the dislocation line traverses is the *slip plane*, as indicated in Figure 1.
- Macroscopic plastic deformation simply corresponds to permanent deformation that results from the movement of dislocations, or slip, in response to an applied shear stress, as represented in Figure 2*a*.

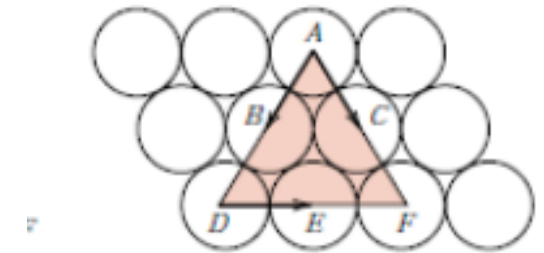
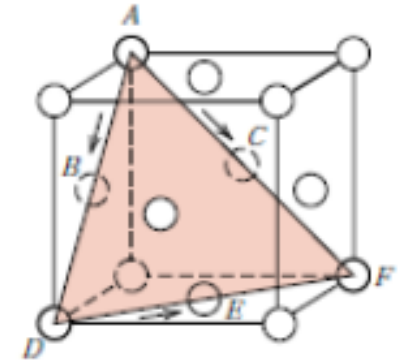
- Dislocation motion is analogous to the mode of locomotion employed by a caterpillar (Figure 3 below).
- The caterpillar forms a hump near its posterior end by pulling in its last pair of legs a unit leg distance. The hump is propelled forward by repeated lifting and shifting of leg pairs.
- When the hump reaches the anterior end, the entire caterpillar has moved forward by the leg separation distance.
- The caterpillar hump and its motion correspond to the extra half-plane of atoms in the dislocation model of plastic deformation.



SLIP SYSTEMS

- Dislocations do not move with the same degree of ease on all crystallographic planes of atoms and in all crystallographic directions.
- Ordinarily there is a preferred plane, and in that plane there are specific directions along which dislocation motion occurs. This plane is called the *slip plane*;
- it follows that the direction of movement is called the *slip direction*. This combination of the slip plane and the slip direction is termed the **slip system**.
- The slip system depends on the crystal structure of the metal and is such that the atomic distortion that accompanies the motion of a dislocation is a minimum.
- For a particular crystal structure, the slip plane is that plane having the most dense atomic packing, that is, has the greatest planar density.
- The slip direction corresponds to the direction, in this plane, that is most closely packed with atoms, that is, has the highest linear density. Planar and linear atomic densities.

- Consider, for example, the FCC crystal structure, a unit cell of which is shown in Figure *a*. There is a set of planes, the 111 family, all of which are closely packed. A (111)-type plane is indicated in the unit cell;
- in Figure *b*, this plane is positioned within the plane of the page, in which atoms are now represented as touching nearest neighbors.
- Slip occurs along 110-type directions within the 111 planes, as indicated by arrows in Figure . Hence, $\{111\}\{110\}$ represents the slip plane and direction combination, or the slip system for FCC.
- Figure *b* demonstrates that a given slip plane may contain more than a single slip direction. Thus, several slip systems may exist for a particular crystal structure; the number of independent slip systems represents the different possible combinations of slip planes and directions.
- For example, for face-centered cubic, there are 12 slip systems: four unique 111 planes and, within each plane, three independent 110 directions.



$\{111\}\{110\}$ slip system ^(b) shown within an FCC unit cell. (b) The (111) plane from (a) and three 110 slip directions (as indicated by arrows) within that plane comprise possible slip systems.

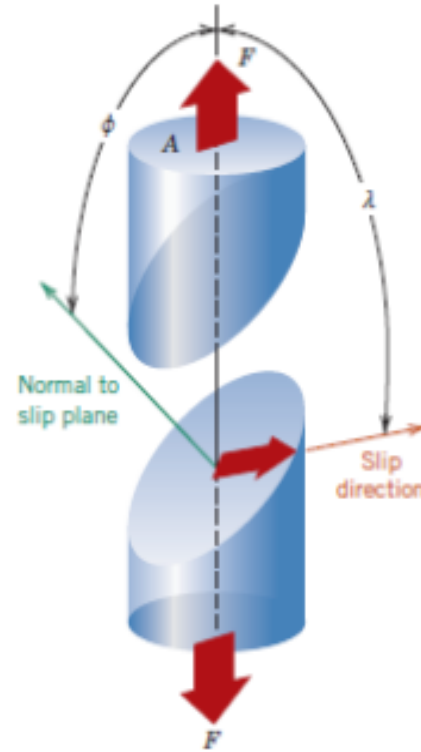
The possible slip systems for BCC and HCP

- The possible slip systems for BCC and HCP crystal structures are listed in Table . For each of these structures, slip is possible on more than one family of planes (e.g., 110, 211, and 321 for BCC).
- For metals having these two crystal structures, some slip systems are often operable only at elevated temperatures.
- Metals with FCC or BCC crystal structures have a relatively large number of slip systems (at least 12). These metals are quite ductile because extensive plastic deformation is normally possible along the various systems.
- Conversely, HCP metals, having few active slip systems, are normally quite brittle.

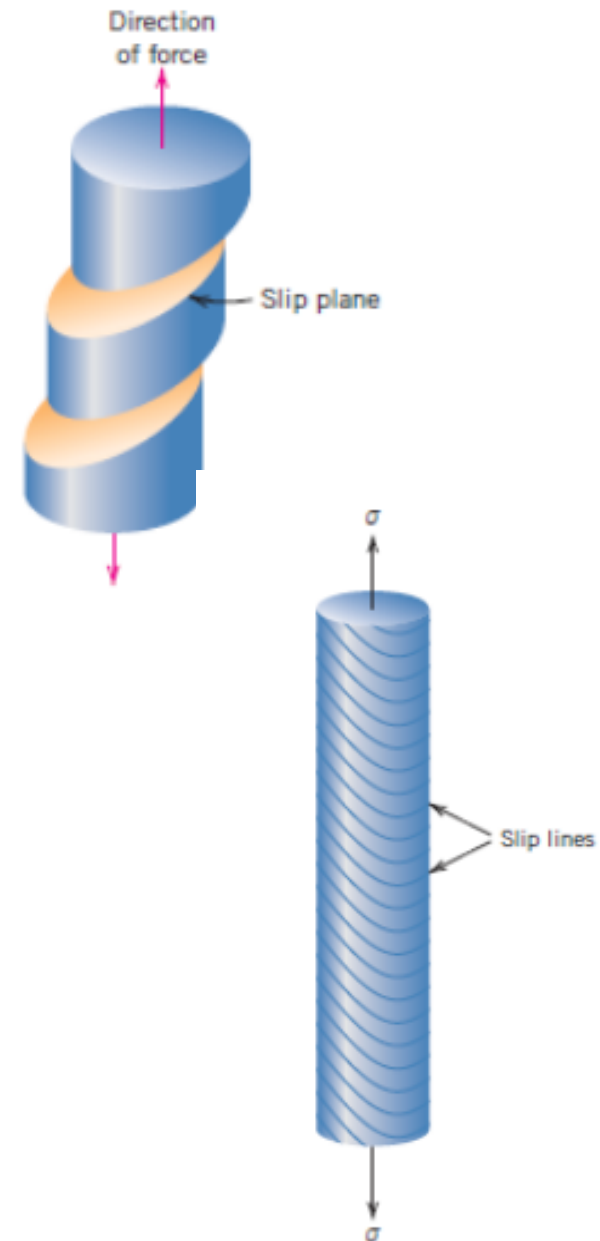
<i>Metals</i>	<i>Slip Plane</i>	<i>Slip Direction</i>	<i>Number of Slip Systems</i>
Face-Centered Cubic			
Cu, Al, Ni, Ag, Au	{111}	$\langle 1\bar{1}0 \rangle$	12
Body-Centered Cubic			
α -Fe, W, Mo	{110}	$\langle \bar{1}11 \rangle$	12
α -Fe, W	{211}	$\langle \bar{1}11 \rangle$	12
α -Fe, K	{321}	$\langle \bar{1}11 \rangle$	24
Hexagonal Close-Packed			
Cd, Zn, Mg, Ti, Be	{0001}	$\langle 11\bar{2}0 \rangle$	3
Ti, Mg, Zr	{10 $\bar{1}$ 0}	$\langle 11\bar{2}0 \rangle$	3
Ti, Mg	{10 $\bar{1}$ 1}	$\langle 11\bar{2}0 \rangle$	6

SLIP IN SINGLE CRYSTALS

- A metal single crystal has a number of different slip systems that are capable of operating.
- The resolved shear stress normally differs for each one because the orientation of each relative to the stress axis (ϕ and λ angles) also differs. However, one slip system is generally oriented most favorably—that is, has the largest resolved shear stress, $\tau R(\max)$:
- $\tau R(\max) = \sigma(\cos \phi \cos \lambda)_{\max}$
- In response to an applied tensile or compressive stress, slip in a single crystal commences on the most favorably oriented slip system when the resolved shear stress reaches some critical value, termed the **critical resolved shear stress** τ_{crss} ;
- it represents the minimum shear stress required to initiate slip and is a property of the material that determines when yielding occurs. The single crystal plastically deforms or yields when $\tau R(\max) = \tau_{crss}$, and the magnitude of the applied stress required to initiate yielding (i.e., the yield strength σ_y) is
- $\sigma_y = \tau_{crss} / (\cos \phi \cos \lambda)_{\max}$
- The minimum stress necessary to introduce yielding occurs when a single crystal is oriented such that $\phi = \lambda = 45^\circ$; under these conditions,
- $\sigma_y = 2\tau_{crss}$



- For a single-crystal specimen that is stressed in tension, deformation is as in Figure 1, where slip occurs along a number of equivalent and most favorably oriented planes and directions at various positions along the specimen length. This slip deformation forms as small steps on the surface of the single crystal that are parallel to one another and loop around the circumference of the specimen as indicated in Figure 1.
- Each step results from the movement of a large number of dislocations along the same slip plane. On the surface of a polished single-crystal specimen, these steps appear as lines, which are called *slip lines*. A schematic depiction of slip lines on a cylindrical specimen that was plastically deformed in tension is shown in Figure 2.
- With continued extension of a single crystal, both the number of slip lines and the slip step width increase. For FCC and BCC metals, slip may eventually begin along a second slip system, the system that is next most favorably oriented with the tensile axis.
- Furthermore, for HCP crystals having few slip systems, if the stress axis for the most favorable slip system is either perpendicular to the slip direction ($\lambda = 90^\circ$) or parallel to the slip plane ($\phi = 90^\circ$), the critical resolved shear stress is zero. For these extreme orientations, the crystal typically fractures rather than deforms plastically.



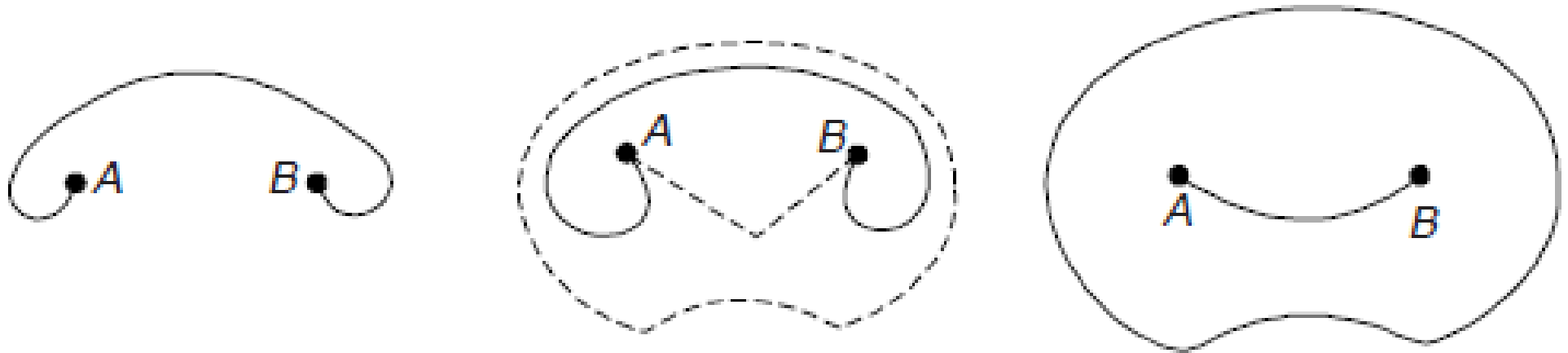
Frank Read Source

- It is observed that slip occurs intensely on a small number of crystal planes and, during the process, some hundreds of dislocations move.
- Obviously, there must be some effective creators or some mechanisms by which these numerous dislocations are produced on a given slip plane.
- These are called as Frank-Read sources.
- A single Frank-Read dislocation source can form hundreds of new dislocations.
- Frank-Read mechanism helps to explain the existence of so called dislocation mills or multiplication of dislocations

- Let us consider a dislocation line, shown by AB in a crystal. We see that the dislocation line or Frank-Read source consists of two nodes A and B .
- In the first situation, when the Burger vector is perpendicular to the line AB , a shear stress parallel to the plane of the figure will exert a force on the dislocation line AB . Due to the action of the shear stress the dislocation line bent outward and produces slip.
- A slip plane usually contains tens of dislocations. For a given stress, the dislocation line AB assume a certain radius of curvature. On further increasing the stress, the dislocation line becomes unstable and expands indefinitely. Figure below illustrates the successive stages.
- Under the action of the applied stress, the fixed dislocation line AB is bent outward until it becomes hemispherical (Fig. (a) and (b)) of radius $AB/2$.
- Note that if the stress is removed at any stage upto this point AB will regain its original shape.



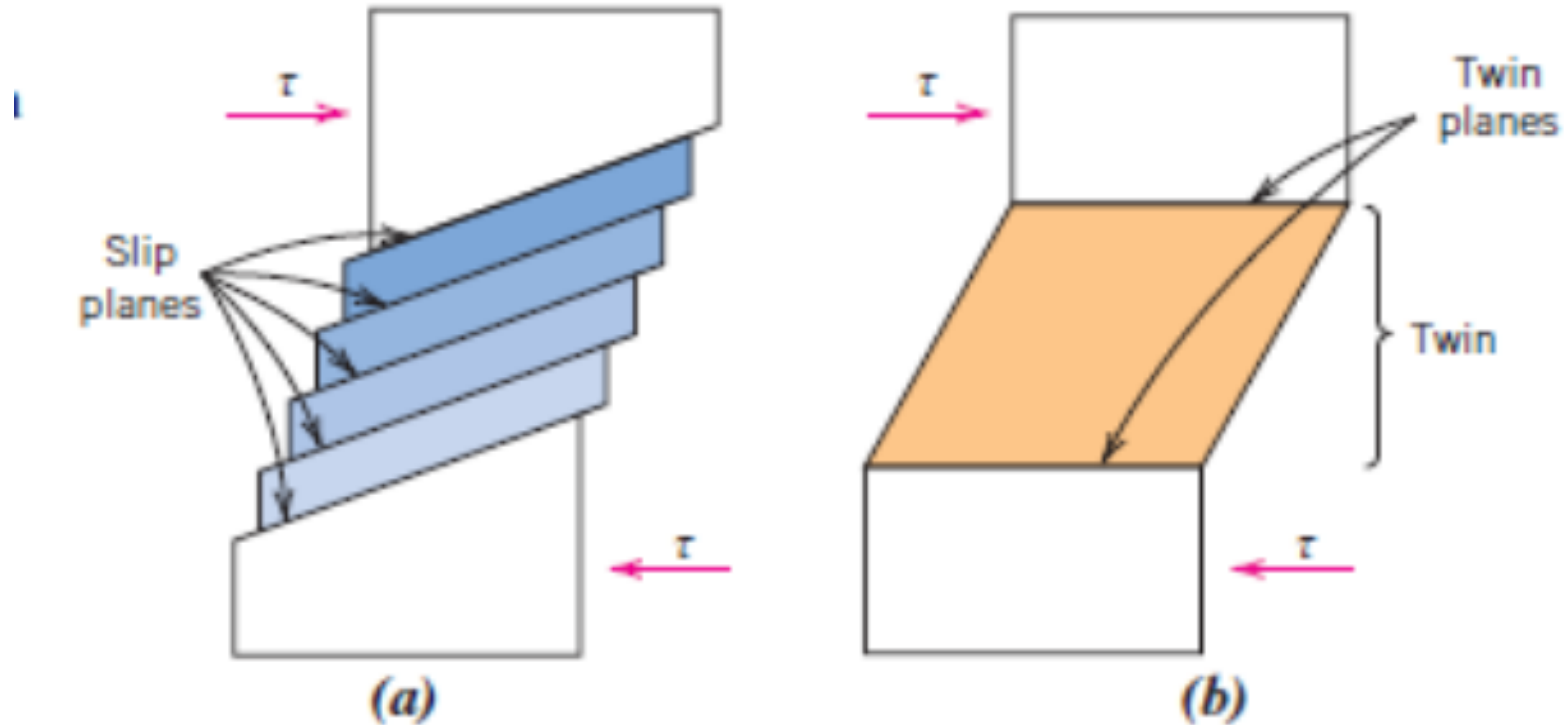
- If the stress is further increased to the stage (Fig. below)) at which the bulge becomes greater than a semicircle, a new system of balance exists and a lower strain energy will be attained by the loop becoming larger (i.e., radius of loop increasing again).
- From that moment on, the bent dislocation propagates spontaneously in the form of two spirals.
- As the spirals meet, they give rise to an expanding dislocation loop and a dislocation section. The dislocation section occupies an initial position and the dislocation source is ready to repeat the cycle.
- A single Frank-Read dislocation source can form hundreds of new dislocations.
- The strengthening effect produced in metals by deformation (strain hardening) is based primarily on an increase of dislocation density.



DEFORMATION BY TWINNING

- In addition to slip, plastic deformation in some metallic materials can occur by the formation of mechanical twins, or *twinning*.
- The concept of a twin is a shear force can produce atomic displacements such that on one side of a plane (the twin boundary), atoms are located in mirror-image positions of atoms on the other side. The manner in which this is accomplished is demonstrated in Figure .
- Blue circles in Figure *b* represent atoms that did not move—red circles those that were displaced during twinning; magnitude of displacement is represented by red arrows. Furthermore, twinning occurs on a definite crystallographic plane and in a specific direction that depend on crystal structure. For example, for BCC metals, the twin plane and direction are (112) and [111], respectively.
- Slip and twinning deformations are compared in Figure for a single crystal that is subjected to a shear stress τ . Slip ledges are shown in Figure *a*; . For twinning, the shear deformation is homogeneous (Figure *b*). These two processes differ from each other in several respects. First, for slip, the crystallographic orientation above and below the slip plane is the same both before and after the deformation; for twinning, there is a reorientation across the twin plane. In addition, slip occurs in distinct atomic spacing multiples, whereas the atomic displacement for twinning is less than the interatomic separation.

For a single crystal subjected to a shear stress τ , (a) deformation by slip; (b) deformation by twinning.

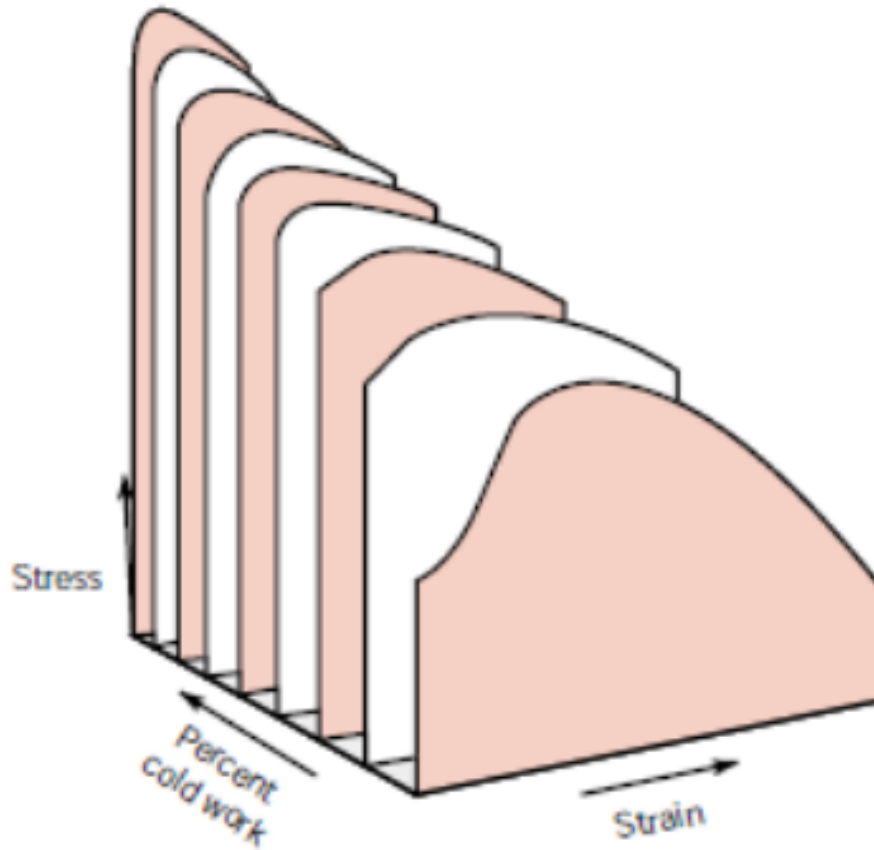


STRAIN HARDENING

- **Strain hardening** is the phenomenon whereby a ductile metal becomes harder and stronger as it is plastically deformed.
- Sometimes it is also called *work hardening*, or, because the temperature at which deformation takes place is “cold” relative to the absolute melting temperature of the metal, **cold working**.
- Most metals strain harden at room temperature.

Strain hardening

- The strain hardening phenomenon is explained on the basis of dislocation–dislocation strain field interactions. The dislocation density in a metal increases with deformation or cold work, due to dislocation multiplication or the formation of new dislocations, takes place .
- Consequently, the average distance of separation between dislocations decreases—the dislocations are positioned closer together. On the average, dislocation–dislocation strain interactions are repulsive. The net result is that the motion of a dislocation is hindered by the presence of other dislocations.
- As the dislocation density increases, this resistance to dislocation motion by other dislocations becomes more pronounced. Thus, the imposed stress necessary to deform a metal increases with increasing cold work.
- Strain hardening is often utilized commercially to enhance the mechanical properties of metals during fabrication procedures. The effects of strain hardening may be removed by an annealing heat treatment.



The influence of cold work
on stress strain behaviour
of carbon steel

End of Lecture

Thanks for your patience

Effect on mechanical properties

Dr. Dinesh Shringi

Contents

- Stress-strain curve.
- Strength and Toughness.
- Hardness, Impact, Ductility and Brittleness.
- Fatigue, Creep, and Fracture.
- Effect of temperature – Recovery, Recrystallization.
- Hot and Cold deformation.

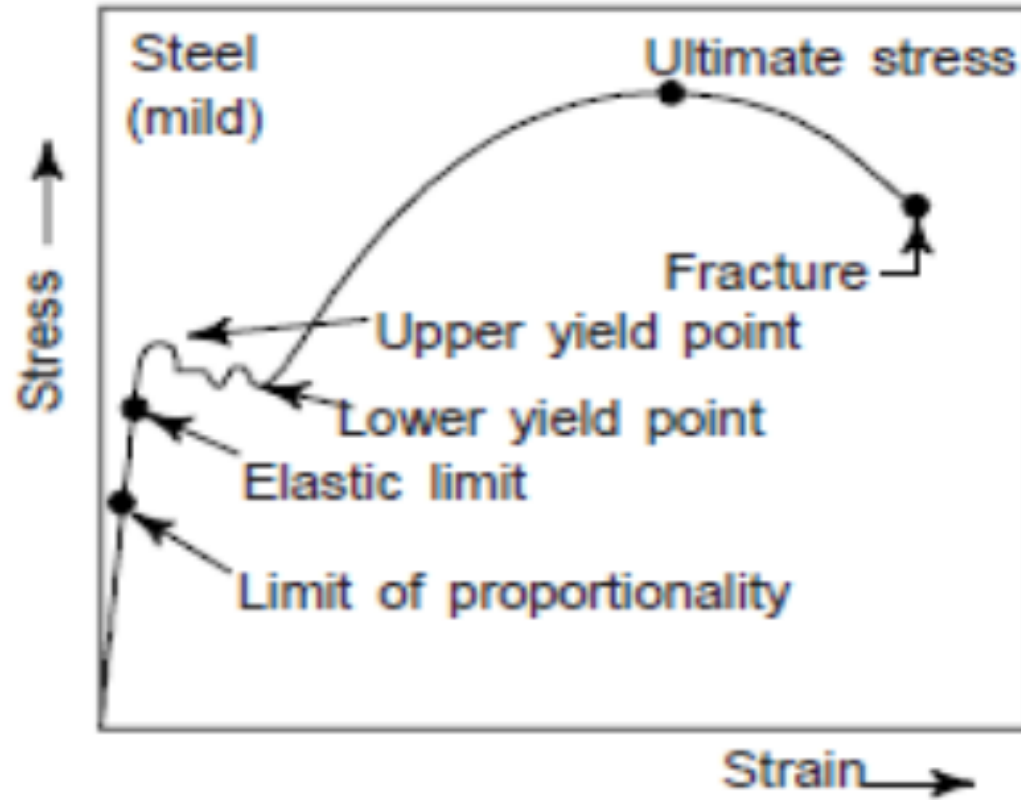
Introduction

- In determining the fabrication and possible practical applications, the mechanical properties of materials, their strength, rigidity and ductility are of vital importance.
- The important mechanical properties of materials are: *elasticity, plasticity, strength, ductility, hardness, brittleness, toughness, stiffness, resilience, malleability, fatigue, creep*, etc.
- The complete specifications of mechanical properties and composition of various materials have been standardised by BIS.
- Materials exhibit a wide range of mechanical properties ranging, e.g., from the ductility of pure copper to the hardness of diamond and most surprising elastic behaviour of rubber.
- There are several materials behave quite differently when stressed in different ways, e.g. steel and wood are stronger in tension whereas cast iron, cement and bricks are much stronger in compression. Obviously, stresses can produce a shape change and may also cause a material to break or fracture. For materials which have to withstand high temperatures, the melting point is also of importance.

Stress-Strain Diagram

- The stress and strain relation is commonly shown by means of a stress-strain diagram.
- These diagrams are obtained by drawing a graph or curve from the data obtained in a tensile test, in which an increasing tensile stress is applied to a specimen .
- There are resulting changes in length which can be observed and recorded by strain measuring devices. Stress-strain for different engineering materials are shown in Fig.

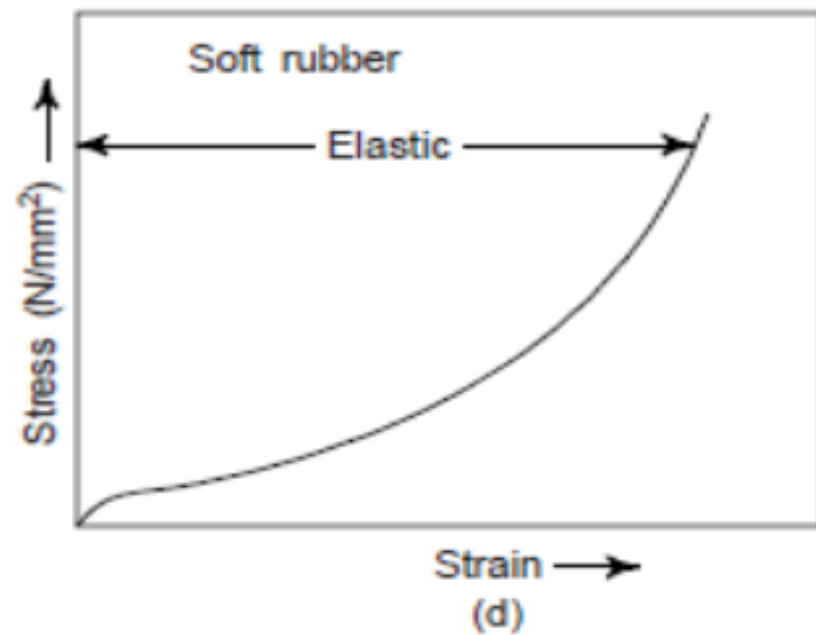
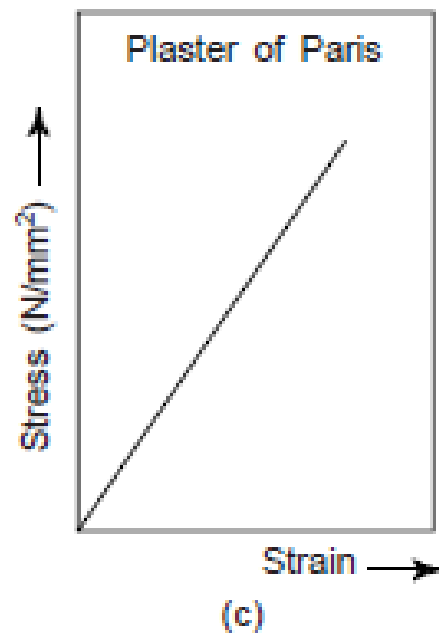
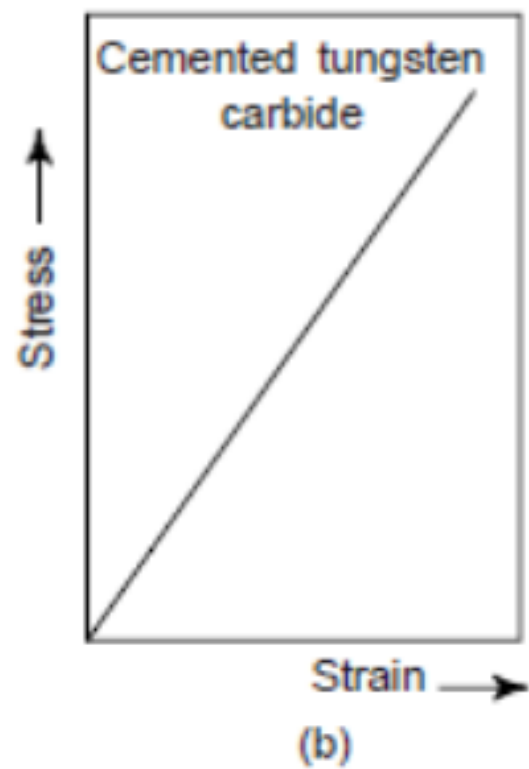
Stress Strain Curve : Mild Steel



(a)

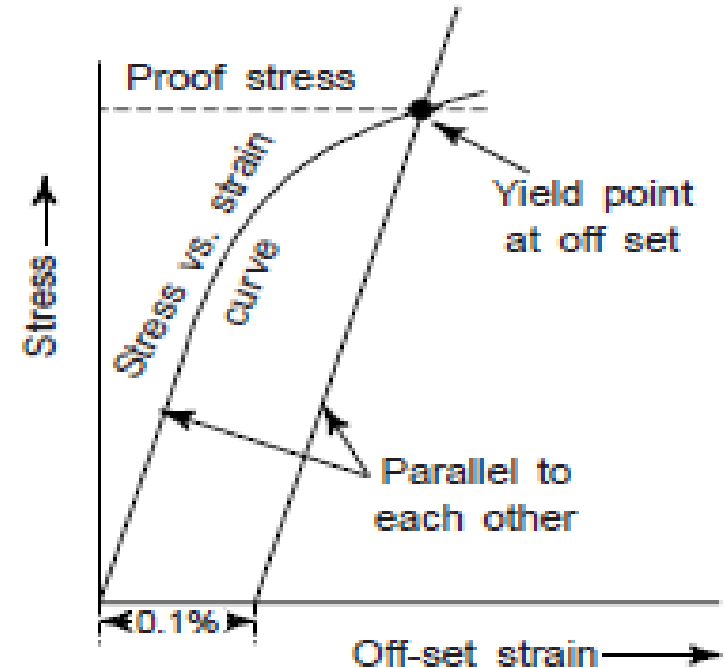
Stress Strain Curve for Mild Materials

- In the case of ductile materials, at the beginning of the test, the material extends elastically. The strain (both longitudinal and lateral) at first increases proportionally to the stress and the sample or specimen returns to its original length on removal of the stress. The limit of proportionality (stress strain) is the stage upto which the specimen, i.e., material obeys Hooke's law perfectly . On further increasing the applied stress, i.e., beyond the elastic limit, it produces plastic deformation so that a permanent extension remains even after the removal of the applied load, i.e. stress. The resultant strain, in this stage begins to increase more quickly than the corresponding stress and continues to increase till the *yield point* is reached.
- We must note that at the yield point the material suddenly stretches. The ratio of applied load to original cross-sectional area is called the *normal stress* and this continues to increase with elongation, due to work hardening or strain hardening, until the tensile stress is maximum.
- This is the value of stress at maximum load and one can calculate it by dividing the maximum load by the original cross-sectional area. This stress is called *ultimate tensile stress* .
- From Fig. it is evident that at a certain value of load the strain continues at slow rate without any further stress or loading. This phenomenon of slow extension increasing with time, at constant stress, is termed *creep*. A neck begins to develop at this point, along the length of the specimen and further plastic deformation is localized within the neck. The cross-sectional area decreases in proportion to the increasing length during elastic elongation. We must note that the volume of the test bar, i.e. specimen remains constant.
- Figure is a stress-strain diagram for mild steel. This diagram clearly shows the limit of proportionality, elastic limit, yield point, ultimate tensile stress and fracture stress at the breaking points. We note that this diagram shows a well-defined yield point.



Stress Strain Curve in Brittle Materials

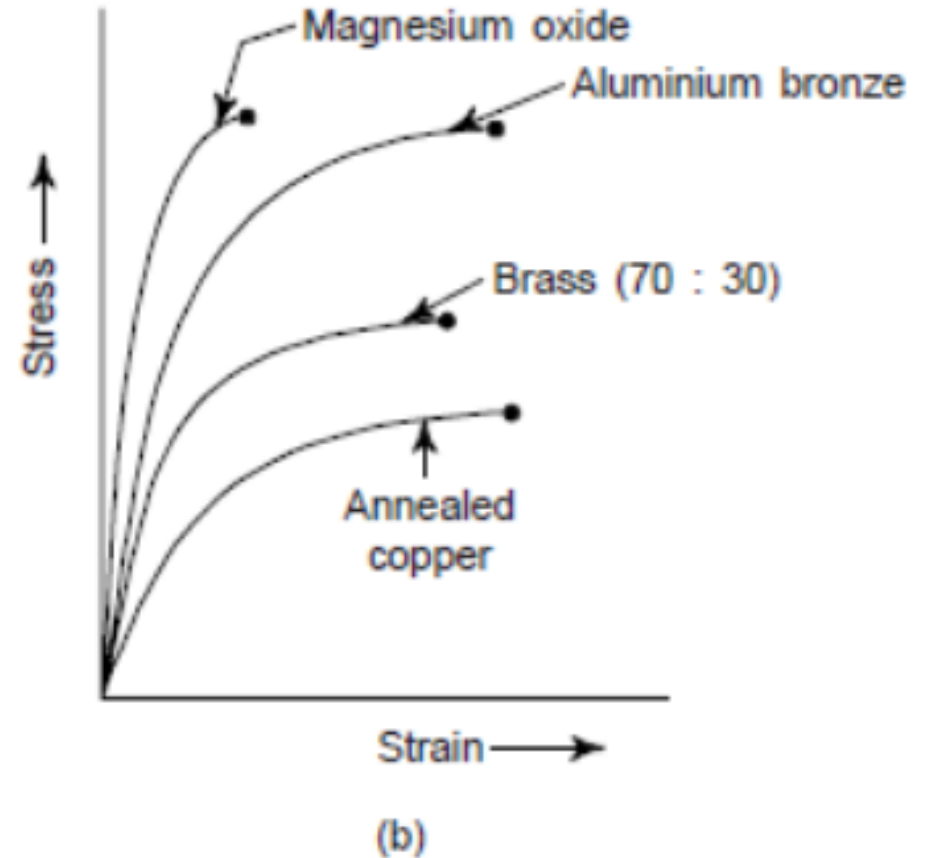
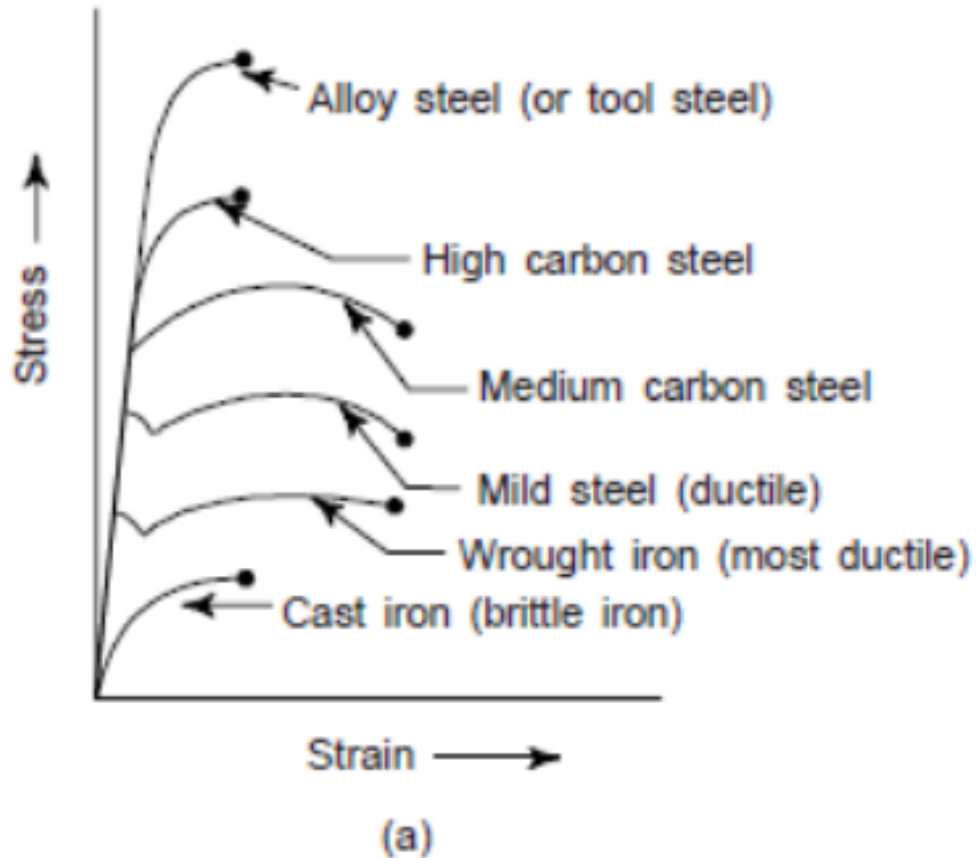
- Poorly defined yield point as in the case of brittle materials is shown in Fig . For the determination of the yield strength in such materials, following the general practice, one has to draw a straight line parallel to the elastic portion of the stress-strain curve at a predetermined strain ordinate value (say 0.1%).
- The point at which this line intersects the stress vs. strain curve is the yield point at off-set and called the yield strength at 0.1% or 0.2% of set strain. In case of hard steels and non-ferrous metals stress is specified corresponding to a definite amount of permanent elongation. This stress is termed as the proof stress.
- the proof stress is applied for 15 seconds and, when removed, the specimen should not lengthen permanently beyond 0.1%. The method of finding the proof stress from the stress-strain curve is shown in Fig.



Some Properties Deduce from Stress Strain Curve

- Stress vs. strain curves also help to explain the properties of ductile materials.
- The greater the angle of inclination of the line of stress vs. strain proportionality to the ordinates, the more elastic is that metal.
- A higher yield point reveals greater hardness of the metal.
- A higher value of the maximum stress point shows that the metal is a stronger one.
- Similarly, the toughness and brittleness of metal are indicated by the distance from the ordinates of the breaking stress or load point.
- The metal is more brittle when the distance is shorter.
- Stress vs. strain curves for ferrous and non-ferrous materials are shown in Fig. (Next Slide) .
- The brittle materials show little or no permanent deformation prior to fracture. Some metals and magnesium oxide exhibit brittle behaviour

Stress-Strain Curve for Ferrous & Non Ferrous Materials



Brittleness:

- It may be defined as the property of a metal by virtue of which it will fracture without any appreciable deformation.
- This property is just opposite to the ductility of a metal. Few examples of brittle materials are cast iron, glass and concrete.
- This property of metals find its importance for design of machine tools, which are subjected to sudden loads.
- Metals with less than 5% elongation are known to be brittle ones.

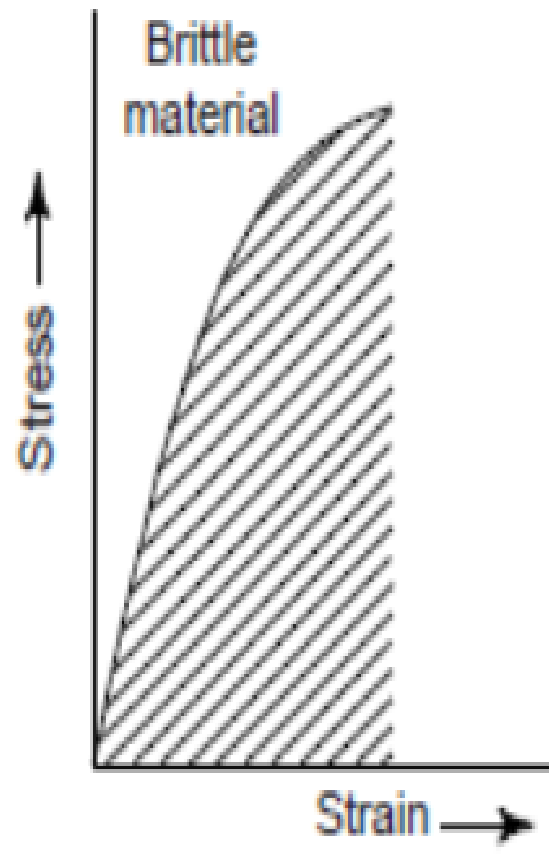
Poisson's ratio (ν)

- When a specimen is stressed by a uniaxial force, it is strained in the direction of the force and also in a direction perpendicular to the direction of the force. The strain in the direction of force is called longitudinal strain and that perpendicular to it as lateral strain. The relationship existing between the lateral strain and axial strain is called Poisson's ratio (ν):
- Poisson's ratio (ν) = Lateral strain / Longitudinal strain = $\epsilon_l / \epsilon_a = \text{constant}$
- Poisson's ratio is an important elastic constant and its value is different for different materials. For most engineering structural materials its value is usually between 0.3 and 0.5. The relation between E , G and ν is

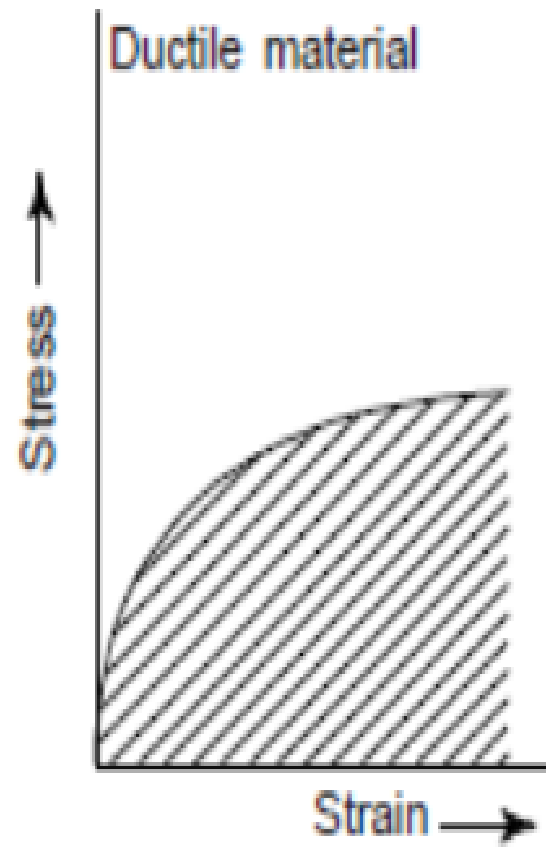
$$\nu = \frac{E}{2G} - 1$$

Toughness

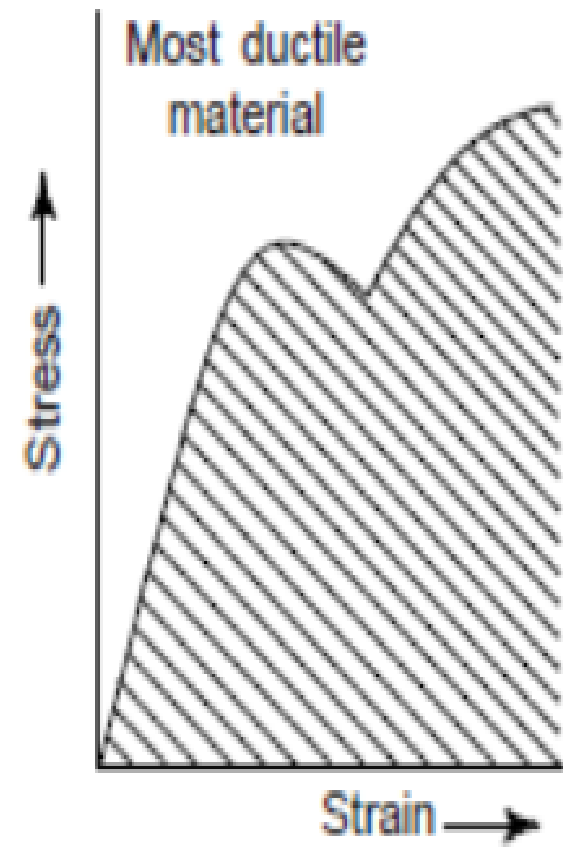
- It may be defined as the property of a metal by virtue of which it can absorb maximum energy before fracture takes place. It is the measurement of ultimate energy strength of material and is expressed as work units/unit volume, i.e. kg fm/m^3 . Toughness is also calculated in terms of area under stress-strain curve (Fig. (A)).
- We may note that the toughness is maximum under Fig. (a) (c). We may note that the total area under stress-strain curve is the work expended in deforming one m^3 of material until it fractures.
- This work is also termed as *modulus of toughness*. Tenacity and hardness of a metal are the measures of toughness.
- The toughness is approximately measured as the product of strength and elongation and these measurements are used for comparison purposes and termed as *merit number*.



(a)



(b)



(c)

Characteristics of Toughness

- Toughness is the property of materials which enables a material to be twisted, bent or stretched under a high stress before rupture.
- The value of toughness falls with the rise in temperature.
- Toughness is highly desirable property for structural and mechanical parts which have to withstand shock and vibration.
- Its importance lies in the selection of a material where the load increases beyond the elastic limit or yield point, e.g. power press punch and pneumatic hammer etc.
- Amongst the tough materials, few are: brass, mild steel, manganese and wrought.
- A ductile material with same strength as non-ductile material requires more energy for breaking and hence is more tough.

Stiffness:

- This may be defined as the property of a metal by virtue of which it resists deformation.
- Modulus of rigidity is the measure of stiffness. The term flexibility is quite opposite of stiffness. The materials which suffer less deformation under load have high degree of stiffness. We may note that the greater the stress required to produce a given strain, the stiffer is the material.
- The stiffness of a metal is of great importance while selecting it for a number or a component of a machine or a structure. To make it clear, let us consider an example of two beams one of steel and other of aluminium. Both the beams are strong enough to carry the given load but aluminium beam will sag or deflect more than the mild steel beam. Obviously, the steel beam is stiffer than the aluminium beam.
- Stiffness is also used in graduating spring balance and spring controlled measuring instruments. Stiffness or flexibility is the function of dimensions, shape and characteristics of the material.

Resilience:

- This may be defined as the property of a metal by virtue of which it stores energy and resists shocks or impacts. It is measured by the amount of energy absorbed per unit volume, in stressing a material upto elastic limit.
- This property is of great importance in the selection of a material used for various types of springs.
- The maximum energy which can be stored in a body upto elastic limit is termed as *proof resilience*.
- Proof resilience per unit volume is termed as *modulus of resilience*. Obviously, the energy stored per unit volume at elastic limit is modulus of resilience.
- The annealed copper is not used for springs due to its very low elastic limit. However, the cold-worked copper has much high elastic limit (and resilience) and therefore it is used for springs. This shows that resilience is associated with high elastic limit. We may note that resilience is of importance for materials required to bear shocks and vibrations.

Endurance:

- This is defined as the property of a metal by virtue of which it can withstand varying stresses (same or opposite nature).
- The maximum value of stress, which can be applied for an indefinite times without causing its failure, is termed as its *endurance* limit.
- For ordinarily steel, the endurance limit is about half the tensile strength.
- This property of a metal is of great importance in the design and production of parts in a reciprocating machines and components subjected to vibrations.
- It is always desirable to keep the working stress of a material well within the elastic limit.
- It is Most Important Property in Designing Movable parts

Hardness

- This is the property of a material (metal) by virtue of it is able to resist abrasion, indentation (or penetration) and scratching by harder bodies. It is the resistance of a material to permanent deformation of the surface.
- In other words, one can define it as the resistance of the metal to penetration by an indenter. The hardness of a surface of the material is, of course, a direct result of interatomic forces acting on the surface of the material. We must note that hardness is not a fundamental property of a material, however, but rather a combined effect of compressive, elastic and plastic properties relative to the mode of penetration, shape of penetrator, etc. Hardness seems to bear a fairly constant relationship to the tensile strength of a given material and thus it can be used as a practical non-destructive test for an approximate idea of the value of that property and the state of the metal near the surface.
- There are several methods of hardness testing, depending either on the direct thrust of some form of penetrator into the metal surface, or on the ploughing of the surface as a stylus is drawn across it under a controlled load, or on the measurement of elastic rebound of an impacting hammer which possessing known energy. Measurements of hardness are the easiest to make and are widely used for industrial design and in research. All hardness tests are made on the surface or close to it. We may note that in mechanical tests the bulk of material is involved.
- Brinell hardness number is the hardness index, calculated by pressing a hardened steel ball (indenter) into test specimen under standard load. Brinell, Rockwell and Vickers hardness tests are used to determine hardness of metallic materials, to check quality level of products, for uniformity of samples of metals, for uniformity of results of heat treatment.

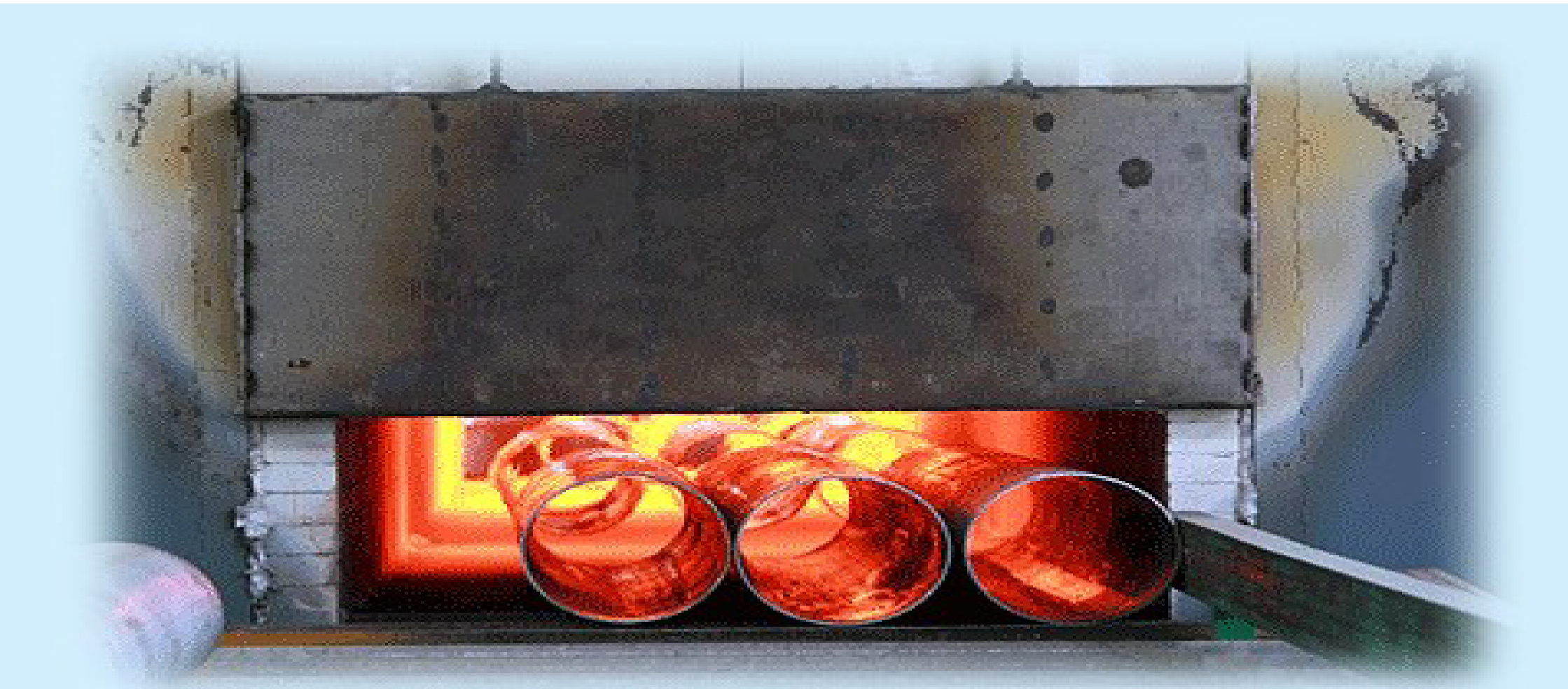
Heat Treatment of Steels

By Dr. Dinesh Shringi

Contents

- Phase transformation,
- TTT diagrams (S curves);
- Stress relief and Annealing,
- Quenching and Hardening,
- Tempering,
- Normalizing,
- Hardenability,
- Case hardening,
- Carburising,
- Nitriding,
- Cyaniding.

Heat Treatment



Heat Treatment

- The amount of carbon present in plain carbon steel has a pronounced effect on the properties of a steel and on the selection of suitable heat treatments to attain certain desired properties.
- Heat treatment may be defined as:
- An operation or combination of operations involving Heating and cooling of a metal/alloy in solid state to obtain desirable

Conditions, e.g., that of relieved stresses

Properties, e.g.,

- Better machinability
- Improved ductility
- Homogeneous structure, etc.

Purpose

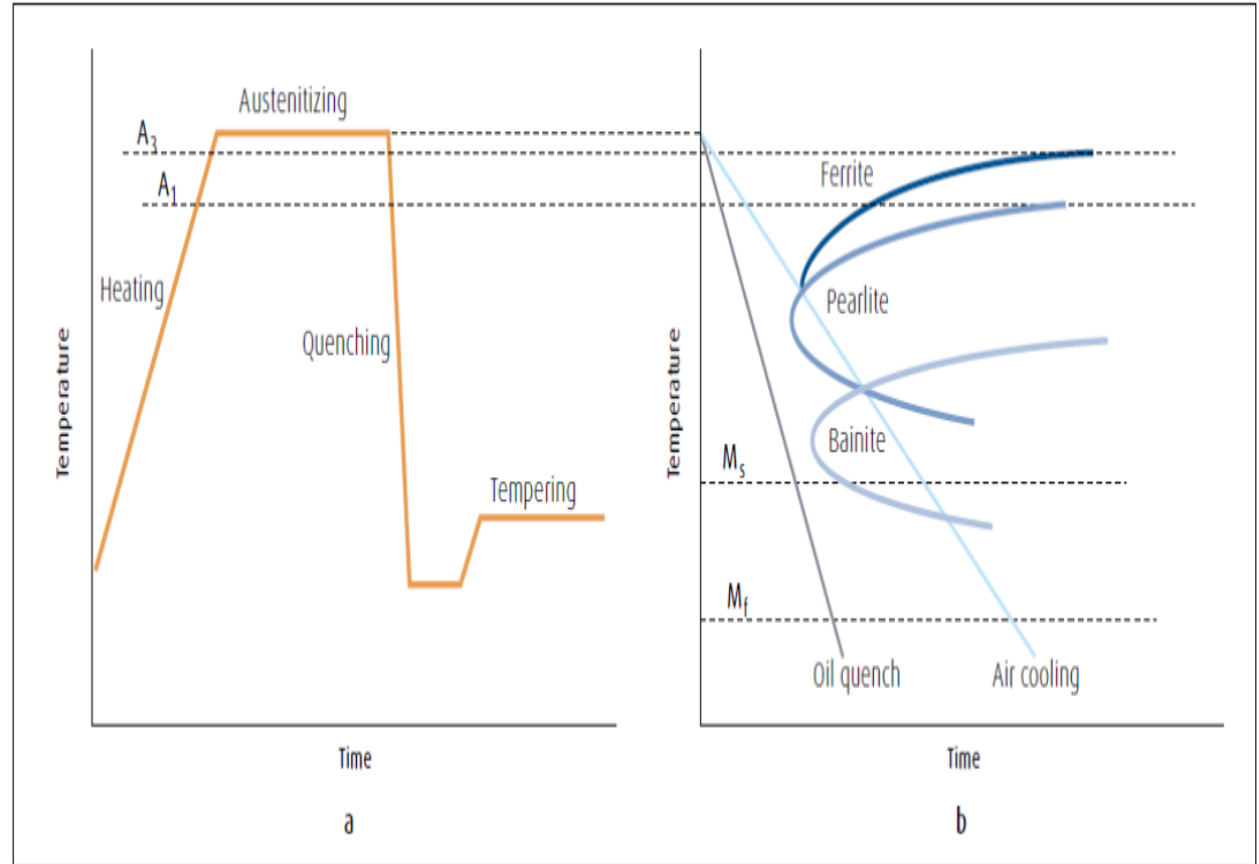
- Cause relief of internal stresses developed during cold working, welding, casting, forging etc.
- Harden and strengthen metals.
- Improve machinability.
- Change grain size
- Soften metals for further (cold) working as in wire drawing or cold rolling.
- Improve ductility and toughness
- Increase , heat, wear and corrosion resistance of materials.
- Improve electrical and magnetic properties.
- Homogenize the structure to remove coring .
- Spheroidize tiny particles, such as those of Fe_3C in steel, by diffusion.

Heat Treatment Theory

- The various types of heat-treating processes are similar because they all involve the heating and cooling of metals.
- they differ in the heating temperatures and the cooling rates used and the final results.
- Ferrous metals (metals with iron) are annealing, normalizing, hardening, and tempering.
- Nonferrous metals can be annealed, but never tempered, normalized, or case-hardened.

Stages of Heat Treatment

- **Stage 1**—Heating the metal slowly to ensure a uniform temperature.
- **Stage 2**—Soaking (holding) the metal at a given temperature for a given period of time.
- **Stage 3**—Cooling the metal to room temperature.



a) Temperature-time cycle and b) its relation to the TTT diagram

- **Soaking:** - Internal structural changes take place. soaking period depends on the chemical analysis of the metal and the mass of the part.
- **Cooling Stage:** - To cool the metal, you can place it in direct contact with a COOLING MEDIUM composed of a gas, liquid, solid, or combination of these.

Soaking Time (hr) = f(Thickness of Metal, Temperature of Heating Medium, and No

Thickness of Metal (inches)	Time of heating to Required Temperature (hr)	Soaking Time (hr)
Up to 1	3/4	1/2
1 to 2	1 1/4	1/2
2 to 3	1 3/4	3/4
3 to 4	2 1/4	1
4 to 5	2 3/4	1
5 to 8	3 1/2	1 1/2

Types

- *Annealing*
- *Normalizing*
- *Hardening*
- Carburizing
- *Tempering*

Annealing

- Steel is annealed to reduce the hardness, improve machine ability, facilitate cold-working, produce a desired microstructure.
- Full annealing is the process of softening steel by a heating and cooling cycle, so that it may be bent or cut easily.
- In annealing, steel is heated above the transformation temperature to form austenite, and cooled very slowly, usually in the furnace.



CLASSIFICATION

- Annealing treatment can be classified into groups based on the following:

1. Heat treatment temperature	2. Phase transformation	3. Specific purpose
• Full annealing • Partial annealing • Sub-critical annealing	• First-order annealing • Second-order annealing	• Full annealing • Isothermal annealing • Diffusion annealing • Partial annealing • Recrystallization annealing • Spheroidisation annealing

Annealing Based on Temperature

- 1.1 In full annealing the steel is heated above the critical temperature(A_3) and then cooled very slowly.
- 1.2 Partial annealing, also known as incomplete annealing or intercritical annealing, involves heating of steel to a temp. lying between lower critical temperature(A_1) and upper critical temperature (A_3 or A_{cm}).
- 1.3 Subcritical annealing is a process in which the maximum temp. to which is heated is always less than the lower critical temperature(A_1).

Classification based on phase transformation features.

- 2.1 **First-order annealing** is performed on steel with the sole aim of achieving some properties.
- Any change in the characteristics of steel achieved by this type of annealing is not correlated to phase transformation.
- it can be performed at a wide range of temperatures above or below the critical temperatures.
- 2.2 The **second-order annealing** differs from the former in the sense that the end results in the former are essentially due to phase transformation which takes place during the treatment.

Types of annealing based on specific purpose

3.1 Full annealing

- In this, steel is heated to its 50°C above the austenitic temperature and held for sufficient time to allow the material to fully form austenite or austenite-cementite grain structure. The material is then allowed to cool slowly so that the equilibrium microstructure is obtained.
- The austenitising temp is a function of carbon content of the steel and can be generalized as:
 - For hypoeutectoid steels and eutectoid steel
 - $A_{c3} + (20-40^{\circ}\text{C})$ [to obtain single phase austenite]
 - For hypereutectoid steels
 - $A_{c1} + (20-40^{\circ}\text{C})$ [to obtain austenite + cementite]

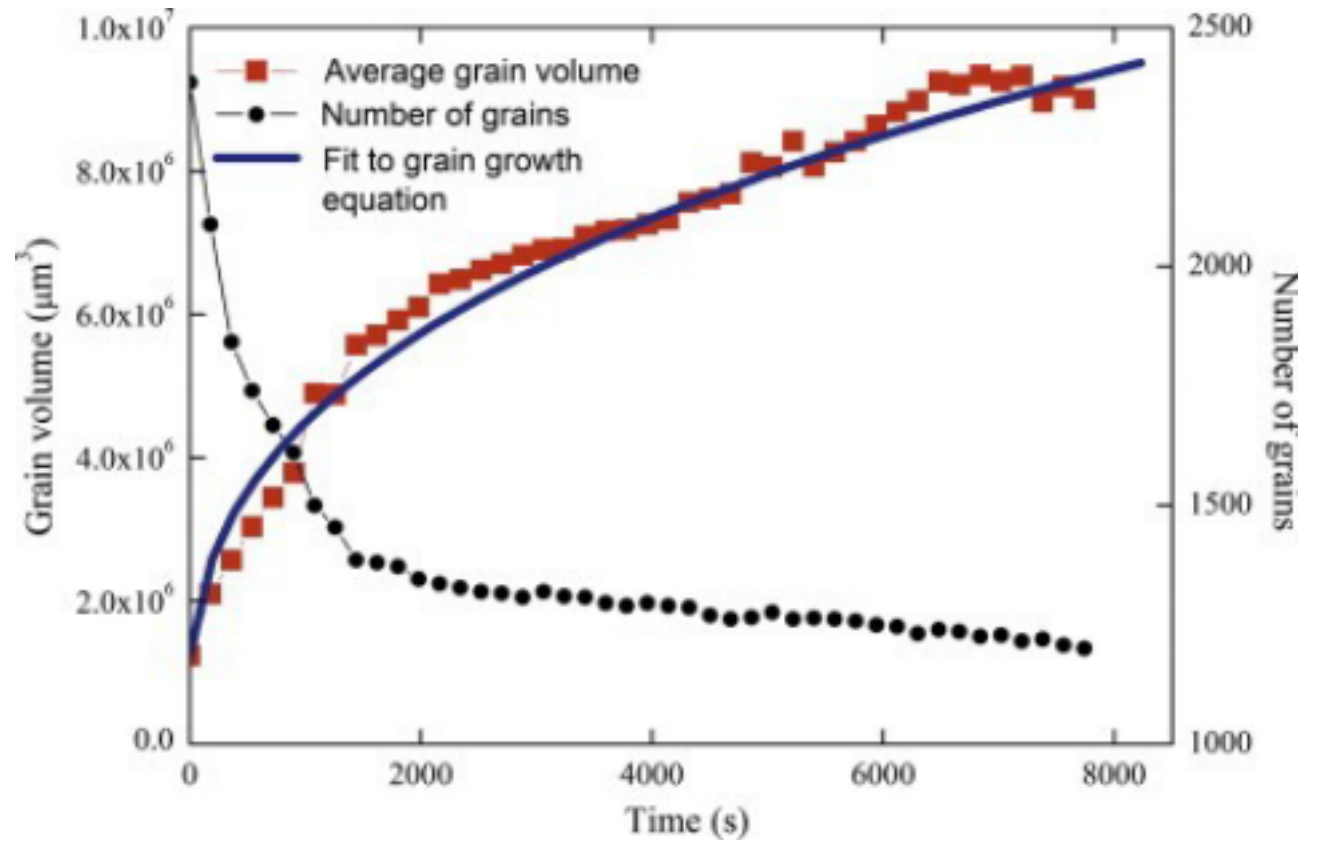
Purpose of full annealing

- To relieve internal stresses
- To reduce hardness and increase ductility
- For refining of grain size
- For making the material having homogeneous chemical composition
- For making the material suitable for high machining processes
- To make steel suitable for undergoing other heat treatment processes like hardening, normalizing etc.

3.2 Isothermal Annealing

- It is a process in which hypoeutectoid steel is heated above the upper critical temperature and this temperature is maintained for a time and then the temperature is brought down below lower critical temperature and is again maintained. Then finally it is cooled at room temperature.
- The prefix “isothermal” associated with annealing implies that transformation of austenite takes place at constant temperature.

The closer the temp of isothermal holding is to A1, coarser is the pearlite, softer is the steel, but longer is the time of isothermal transformation.



- Advantages:

- Improved machinability.
- Homogeneous structure and better surface finish.
- Time required for complete cycle is comparably less.
- The process is of great use for alloy steels as the steels have to be cooled slowly.

- Limitation:

- It is suitable only for small-sized components.
- Heavy components cannot be subjected to this treatment because it is not possible to cool them rapidly and uniformly to the holding temperature at which transformation occurs. Thus structure won't be homogeneous. so mechanical properties will vary across the cross-section.

3.3 Diffusion Annealing

- This process, also known as homogenizing annealing, is employed to remove any structural non-uniformity like dendrites, columnar grains and chemical inhomogeneity which promote brittleness and reduce ductility and toughness of steel.
- Process:
 - Steel is heated sufficiently above the upper critical temperature (say, 1000-2000°C), and held at this temperature for 10-20 hours, followed by slow cooling.
 - Segregated zones are eliminated, and a chemically homogeneous steel is obtained by this treatment as a result of diffusion.
 - Heating to such a high temp. results in considerable coarsening of austenitic grains & heavy scale formation. The coarse austenite thus obtained further transforms to coarse pearlite on cooling, which is not a desirable structure as mechanical properties are impaired.

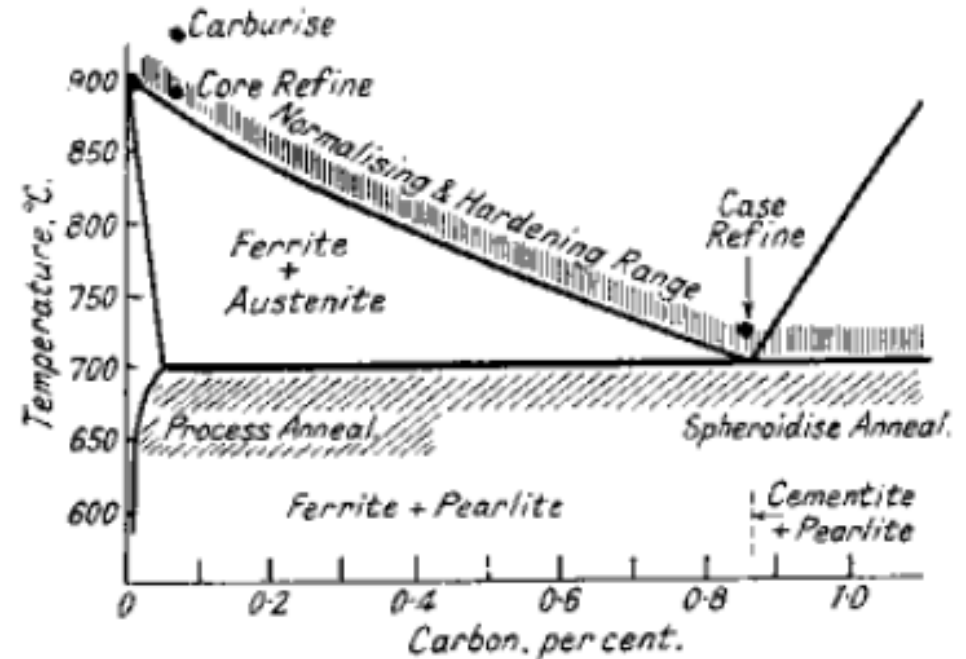
- The main aim of homogenising annealing is to make the composition uniform, i.e to remove chemical heterogeneity.
- The impact energy and ductility of the steel increase as the homogenizing temperature increases and the hardness, yield strength and tensile strength decrease with an increase in the homogenizing temperature.
- Homogenizing annealing has a few shortcomings as well. It results in:
 - Grain coarsening of austenite, thereby impairing the properties
 - Thick scales on the surface of steels
 - It is an expensive process

3.4 Partial Annealing

- Partial annealing, also known as inter-critical annealing or incomplete annealing, is a process in which steel is heated between A_1 and A_{cm} and is followed by slow cooling.
- Generally, hypereutectoid steels are subjected to this treatment. The resultant microstructure consists of fine pearlite and cementite instead of coarse pearlite and a network of cementite at grain boundaries.
- As low temperatures are involved in this process, it is less expensive than full annealing.
- Hypoeutectoid steels are subjected to this treatment to improve machinability.
- However, steels with coarse structure of ferrite and pearlite are not suitable for this treatment. This is because only a considerable amount of ferrite remains untransformed, and only a part of it along with pearlite transforms to austenite.
- This coarse or acicular untransformed ferrite results in poor mechanical properties.

3.5 Recrystallization Annealing

- The process consists of heating steel above the recrystallization temperature, holding at this temperature and cooling thereafter.
- It is used to treat work-hardened parts made out of low-Carbon steels ($< 0.25\%$ Carbon).
- This allows the parts to be soft enough to undergo further cold working without fracturing.



- Recrystallization temp(T_r) is given by:

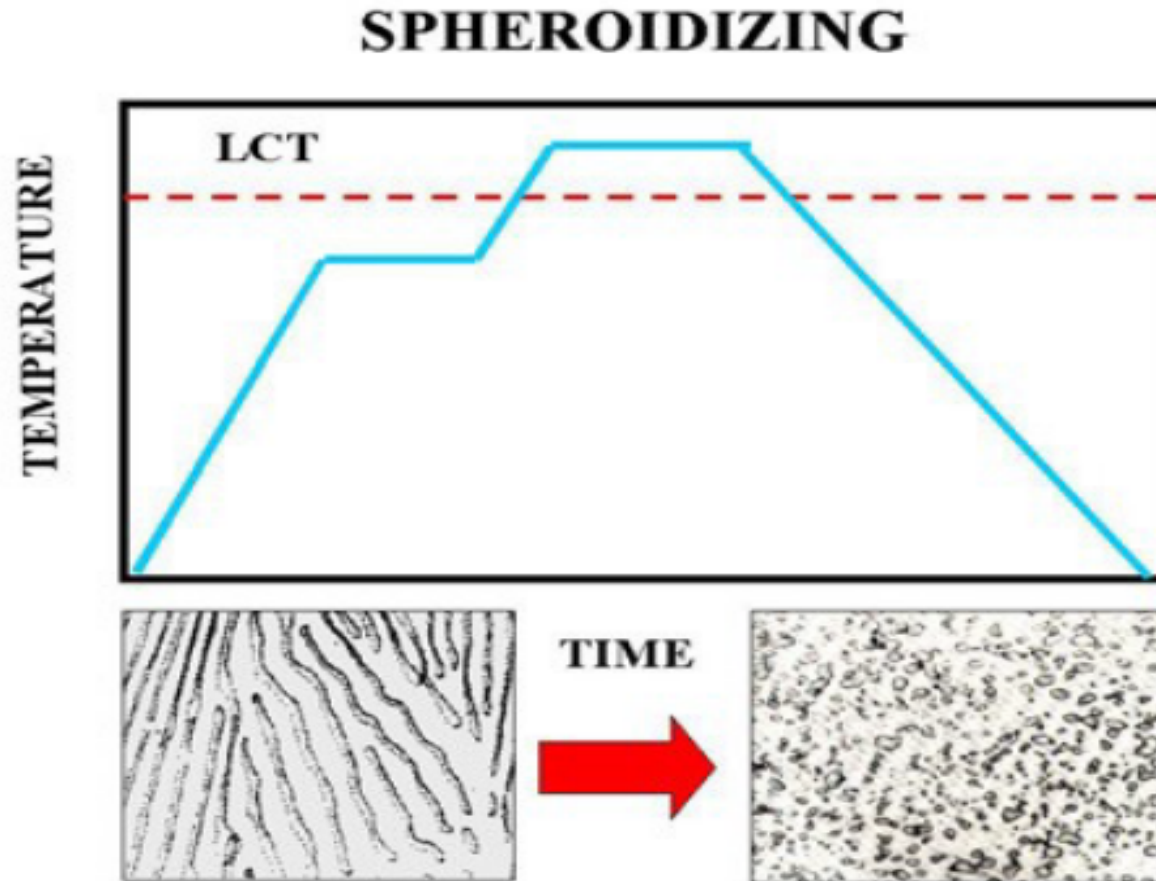
$$T_r = (0.3-0.5)T_{mp} \quad (T_{mp} \text{ is Melting point temperature})$$

- As little scaling and decarburization occurs in recrystallization annealing, it is preferred over full annealing.
- No phase change takes place and the final structure consists of strain-free, equiaxed grains of fine ferrite produced at the expense of deformed elongated ferrite grains.
- However It would produce very coarse grains if the steel has undergone critical amount of deformation. In such cases, full annealing is preferred.
- Aims of Recrystallization Annealing
 - To restore ductility
 - To refine coarse grains
 - To improve electrical and magnetic properties in grain-oriented Si steels.

3.5 Spheroidization annealing

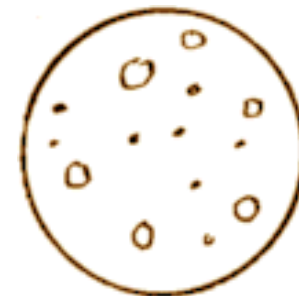
- Spheroidization annealing consists of heating, soaking and cooling, invariably very slowly to produce spheroidal pearlite or globular form of carbides in steels.
- To improve the machinability of the annealed hypereutectoid steel spheroidize annealing is applied.
- Hypereutectoid steels consist of pearlite and cementite. The cementite forms a brittle network around the pearlite. This presents difficulty in machining the hypereutectoid steels.
- This process will produce a spheroidal or globular form of a carbide in a ferritic matrix which makes the machining easy.
- Prolonged time at the elevated temperature will completely break up the pearlitic structure and cementite network. The structure is called spheroidite.

Spheroidizing process applied at a temperature below and above the LCT.



Spheroidising Process:

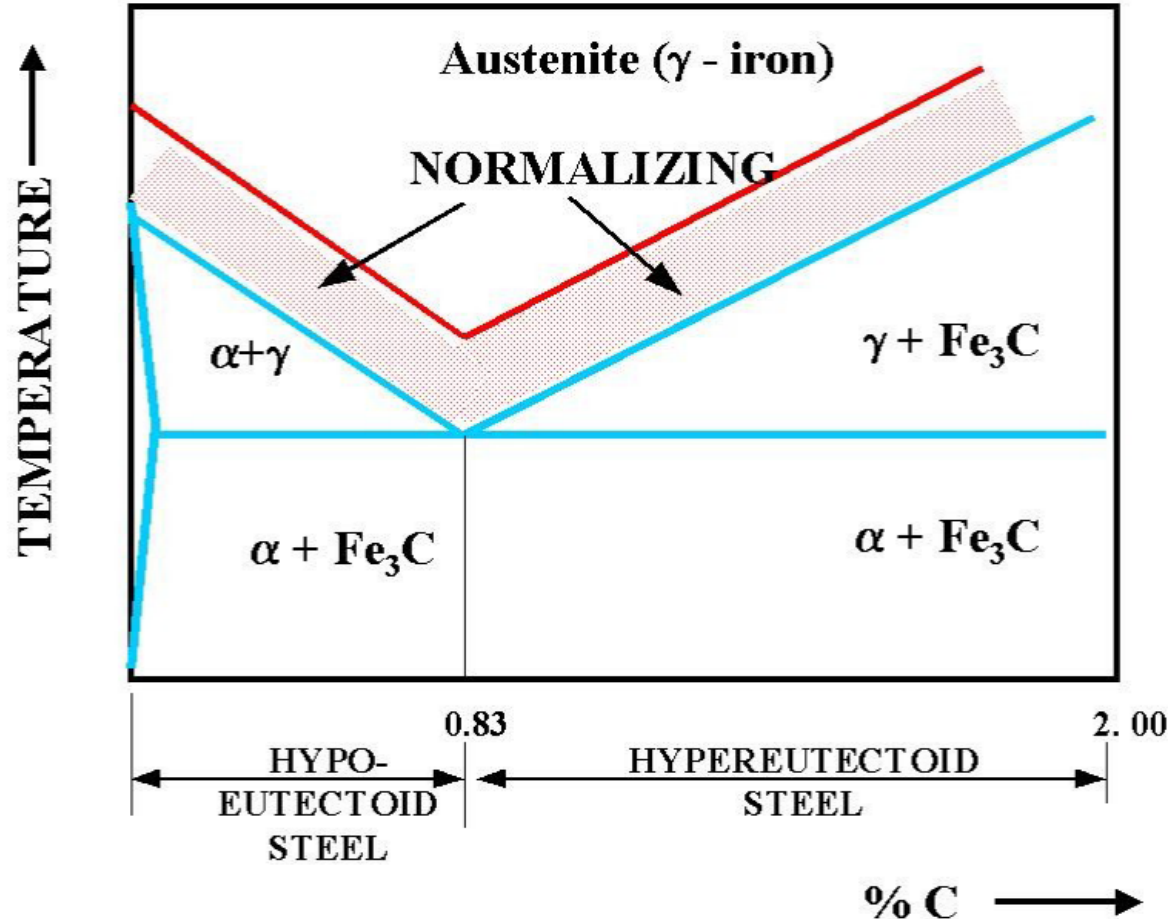
- Heat the part to a temperature just below the Ferrite- Austenite line, line A1 or below the Austenite- Cementite line, essentially below the 727 °C (1340°F) line. Hold the temperature for a prolonged time and follow by fairly slow cooling. Or
- Cycle multiple times between temperatures slightly above and slightly below the 727 °C (1340 °F) line, say for example between 700 and 750 °C (1292 -1382 °F), and slow cool. Or
- For tool and alloy steels heat to 750 to 800 °C (1382- 1472 °F) and hold for several hours followed by slowcooling.
- All these methods result in a structure in which all the Cementite is in the form of small globules (spheroids) dispersed throughout the ferrite matrix.
- This structure allows for improved machining in continuous cutting operations such as lathes and screw machines. Spheroidization also improves resistance to abrasion.
- **Aims of Spheroidization Annealing:**
 - minimum hardness
 - maximum ductility
 - maximum machinability
 - maximum softness



NORMALIZING

- The normalizing of steel is carried out by heating above the UCT (Upper Critical Temperature) to single phase austenitic region to get homogeneous austenite, soaking there for some time and then cooling it in air to room temperature.
- The austenitising temperature range are:
 - For hypoeutectoid steels and eutectoid steel
 $A_{c3} + (40-60\text{ C})$
 - For hypereutectoid steels
 $A_{cm} + (30-50\text{ C})$
- During normalising we use grain refinement which is associated with allotropic transformation upon heating $\gamma \rightarrow \alpha$
- Parts that require maximum toughness and those subjected to impact are often normalized.
- When large cross sections are normalized, they are also tempered to further reduce stress and more closely control mechanical properties.
- The microstructure obtained by normalizing depends on the composition of the castings (which dictates its hardenability) and the cooling rate.

Figure below shows the normalizing temperatures for hypoeutectoid and hypereutectoid steels



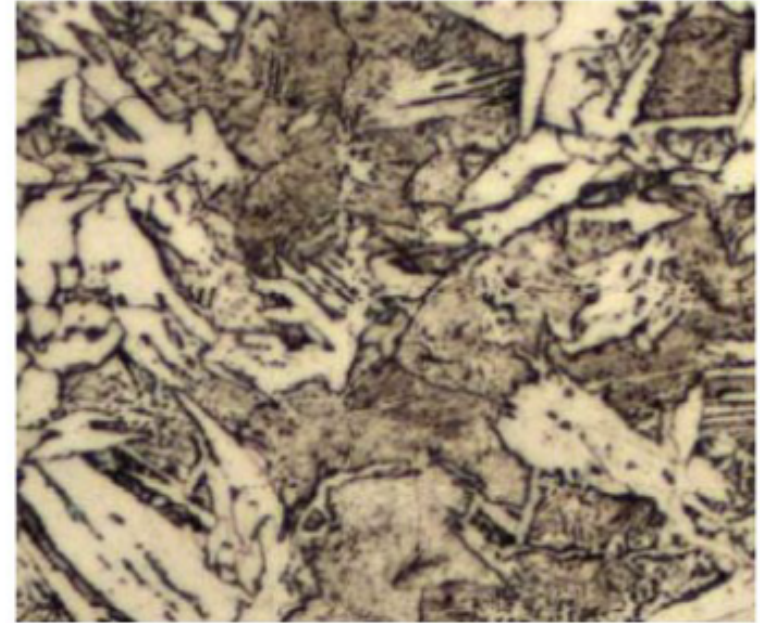
AIMs OF NORMALIZING

- To produce a harder and stronger steel than full annealing
- To improve machinability
- To modify and/or refine the grain structure
- To obtain a relatively good ductility without reducing the hardness and strength
- Improve dimensional stability
- Produce a homogeneous microstructure
- Reduce banding
- Provide a more consistent response when hardening or case hardening

EFFECT OF SOAKING TIME ON THE MICROSTRUCTURE:



(a)



(b)

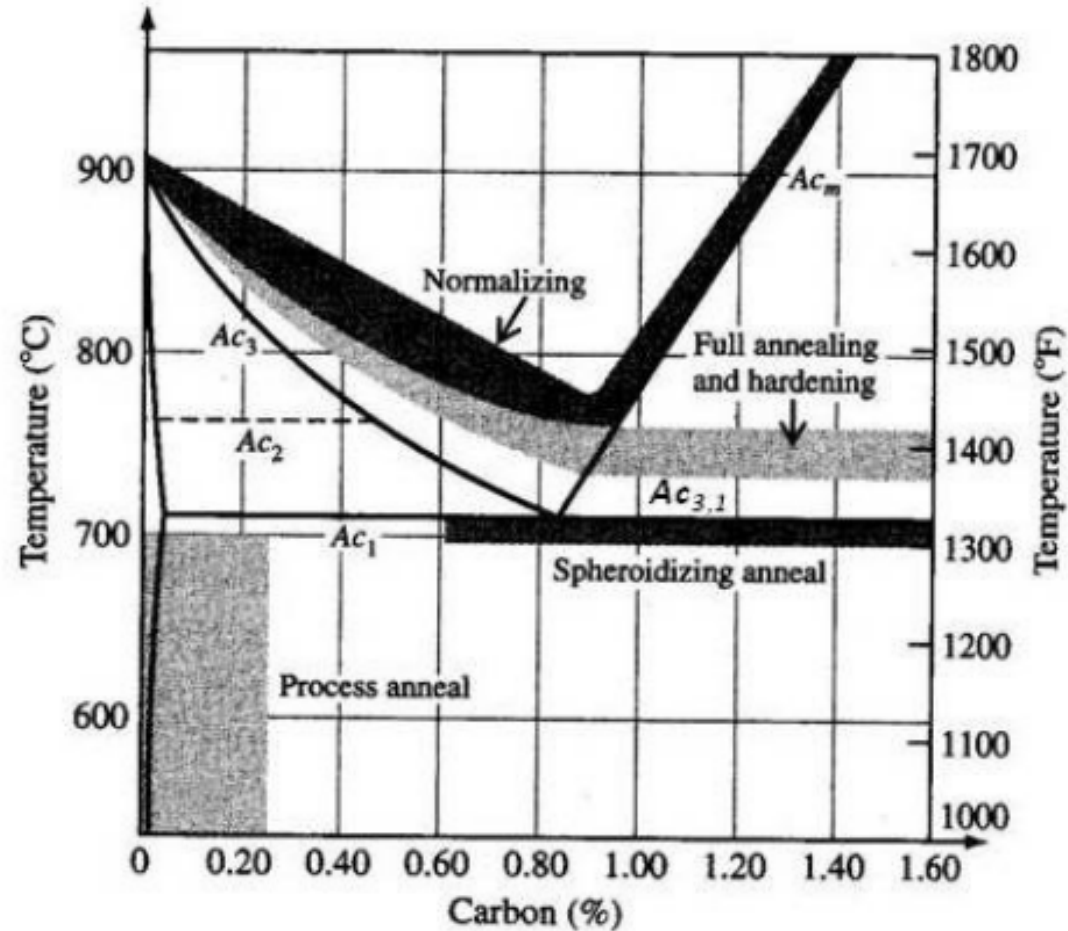
50 μm

Acicular ferrite in mixture with polygonal ferrite and pearlite, normalized at (a) 960°C, 10 minutes soaking time (b) 960°C, 40 minutes soaking time.

COMPARISON OF ANNEALING AND NORMALIZING

- The metal is heated to a higher temperature and then removed from the furnace for air cooling in normalizing rather than furnace cooling.
- In normalizing, the cooling rate is slower than that of a quench-and temper operation but faster than that used in annealing. As a result of this intermediate cooling rate, the parts will possess a hardness and strength somewhat greater than if annealed.
- Fully annealed parts are uniform in softness (and machinability) throughout the entire part; since the entire part is exposed to the controlled furnace cooling. In the case of the normalized part, depending on the part geometry, the cooling is non-uniform resulting in non-uniform material properties across the part.
- Internal stresses are more in normalizing as compared to annealing.
- Grain size obtained in normalizing is finer than in annealing.
- Normalizing is a cheaper and less time-consuming process.

Comparison of temperature ranges in annealing and normalizing



Temperature Range for Heat Treatment of Carbon Steels

